



Submersible Borehole Pump

UPA S

Size: UPA S 300

Nominal Pump Diameter: 12 Inches

Type Series Booklet



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Type Series Booklet UPA S

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Water Supply

Submersible Borehole Pumps

UPA S 300



Main applications

- Spray irrigation systems
- General irrigation systems
- Pressure boosting
- Groundwater management
- Water supply systems
- Offshore processes

Fluids handled

- Drinking water
- River water, lake water and groundwater
- Maximum sand content in fluid handled depending on particle size and duration, sand content up to 250 g/m³¹⁾

Operating data

Table 1: Operating properties

Characteristic		Value		
		50 Hz	60 Hz	100 Hz
Flow rate	Q [m ³ /h]	≤ 542	≤ 655	≤ 561
	Q [l/s]	≤ 151	≤ 182	≤ 156
Head	H [m]	≤ 395	≤ 399	≤ 423
Fluid temperature	T [°C]	≤ 50	≤ 50	≤ 50
Speed	n [rpm]	≤ 2900	≤ 3500	= 3000
Well diameter	D [mm]	≥ 300	≥ 300	≥ 300
	D ["]	≥ 12	≥ 12	≥ 12

Design details

Design

- Centrifugal pump
- Single-stage or multistage
- Single-suction
- Ring-section design
- Rigid connection between pump and motor

Type of installation

- Vertical installation
- Horizontal installation
- Angled installation

Drive

Asynchronous motor:

- With squirrel cage motor for submerged use
- Standard connection to NEMA
- Frequency 50 Hz/60 Hz
- IP68 enclosure
- DOL starting or star-delta starting
- Frequency of starts ≤ 15 starts per hour
- Winding J1 (PPC/HDPE) or, for higher temperatures, J2 (radiation-crosslinked PE2/PA)

Synchronous motor:

- Interior (buried) permanent magnet synchronous motor (IPMSM)
- IP68 enclosure
- Standard connection to NEMA
- Frequency 100 Hz
- Winding J2 (radiation-crosslinked PE2/PA)
- Frequency of starts ≤ 15 starts per hour

Electrical connection

- Delivered ex-factory with 1 or 2 motor leads (including earth conductor and internal earthing)
- Extension cable connected with water-tight cable connector
- Motor lead and extension cable suitable for drinking water use

Impeller type

- Mixed flow design

Bearings

- Radial plain bearings
- Pump bearings lubricated by fluid handled; motor bearings lubricated by water fill
- Axial thrust is balanced by a tilting-pad thrust bearing in the motor (lower end)

¹ For 50 g/m³ and above, the more wear-resistant version with silicon carbide bearings is recommended.

Connections

- Pump screw-ended or flanged
- Threaded end to DIN ISO 228, Part 1 (up to 6 inch)
- NPT threaded end to ASME B1.20.1-2013
- Flanged end to DIN EN 1092
- With lift check valve or connection nozzle

Designation

Table 2: Designation example

Position																																
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	
U	P	A	S		3	0	0	-	2	7	0	/	0	1	C	C	+	U	M	A	S	2	0	0	-	7	5	/	4	2	C	
Pump																	+	Motor														
Pump set																																

Table 3: Designation key

Position	Code	Description
1-5	Type series	
	UPA S	Precision (investment) cast variant
6-8	Nominal pump diameter [mm]	
	300	
9-12	Flow rate at best efficiency point [m³/h]	
	-270	270
	-370	370
13	/	
14-15	Number of stages	
	01, 02, 03, 04, 05, 06, 07, 08, 09	
16	Pump material	
	D	Duplex 1.4517
	C	Stainless steel AISI 316
17	Impeller material	
	D	Duplex 1.4517
	C	Stainless steel AISI 316
18	+	With motor
	-	Without motor
19-22	Motor version	
	UMA	Asynchronous motor
	UMA S	Permanent magnet synchronous motor
23-25	Nominal motor diameter [mm]	
	200, 250, 300	
26	-	
27-28	Motor rating PN [kW]	
	37	37
	...	...
	400	400
29	/	
30	Number of motor poles	
	2	2 poles
	4	4 poles
31	Winding insulation	
	1	J1 (PPC/HDPE)
	2	J2 (radiation-crosslinked PE2/PA)
32	Motor material	
	D	1.4539 / AISI 904L
	C	Stainless steel AISI 316
	G	Grey cast iron

Materials

Table 4: Pump material selection, UPA S 300

Component	Material variant	
	CC	DD
Impeller	CrNiMo steel (1.4408)	CrNiMo steel (1.4517)
Suction casing	CrNiMo steel (1.4408)	CrNiMo steel (1.4517)
Screws, bolts and nuts	CrNiMo steel (A4)	CrNiMo steel (1.4462)
Stage casing	CrNiMo steel (1.4408)	CrNiMo steel (1.4517)
Shaft	CrNiMo steel (1.4462)	CrNiMo steel (1.4462)

Table 5: Motor material selection, UMA 200, UMA-S 200, UMA 250, UMA-S 250, UMA 300

Component	Size	Material variant		
		G	C	D
Casing	UMA 200, UMA-S 200	Grey cast iron (EN-GJL-200)	CrNiMo steel (1.4408)	CrNiMo steel (1.4539)
	UMA 250, UMA-S 250	Grey cast iron (EN-GJL-200)	CrNiMo steel (1.4408)	CrNiMo steel (1.4539)
	UMA 300	Grey cast iron EN-GJL-250	-	CrNiMo steel (1.4517)
Screws, bolts and nuts	UMA 200, UMA-S 200	CrNiMo steel (A4-70)	CrNiMo steel (A4-70)	CrNiMo steel (1.4539)
	UMA 250, UMA-S 250	CrNiMo steel (A4-70)	CrNiMo steel (A4-70)	CrNiMo steel (1.4539)
	UMA 300	CrNiMo steel (1.4571)	-	CrNiMo steel (1.4462)
Shaft	UMA 200, UMA-S 200	CrNiMo steel (1.4460)	CrNiMo steel (1.4460)	CrNiMo steel (1.4462)
	UMA 250, UMA-S 250	CrNiMo steel (1.4460)	CrNiMo steel (1.4460)	CrNiMo steel (1.4462)
	UMA 300	CrNiMo steel (1.4462-C45+N)	-	CrNiMo steel (1.4462-C45+N)

Product benefits

- High efficiencies through optimised, precision (investment) cast hydraulic design
- High operating reliability as all components are made of high-quality stainless steel.
- Overall efficiency can be increased further by combining the pump with an UMA S synchronous motor with ratings of up to 250 kW
- Reliable operation by anti-jam lift check valve
- Easy to install with service-friendly hydraulic system design
- Solids-induced wear can be minimised by metal wear rings and bearings with enhanced wear resistance (optional).

Certifications

Table 6: Overview

Label	Effective in:	Comment
	All countries	Certified quality management to ISO 9001
	France	Approved in accordance with the French drinking water regulation

Product information as per Regulation No. 1907/2006 (REACH)

For information as per European chemicals regulation (EC) No. 1907/2006 (REACH) see <https://www.ksb.com/en-global/company/corporate-responsibility/reach>.

Overview of submersible motors

The following submersible motors are available:

Table 7: Selection table

Well diameter	Motor size	Voltage range	Rated power
8 inches	UMA 200	Up to 1 kV	37 to 90 kW
8 inches	UMA-S 200	400 V	45 to 130 kW
10 inches	UMA 250	Up to 1 kV	85 to 190 kW
10 inches	UMA-S 250	400 V	150 to 250 kW
12 inches	UMA 300	Up to 1 kV	250 to 400 kW

- Further motor data for the UMA asynchronous motor see type series booklet No. 3455.51
- Further motor data for the UMA-S synchronous motor see type series booklet No. 3455.52

Technical data, 50 Hz

Information on characteristic curves

The characteristic curves shown are intended to allow pre-selection. Refer to the quotation for the precise selection data.

Tolerance to ISO 9906 Cl. 2B

The characteristic curves shown are valid for:

- Water at 20 °C
- Density 998 kg/m³
- Viscosity 1.00 mm²/s

Selection chart UPA S 300, 50 Hz, n = 2900 rpm

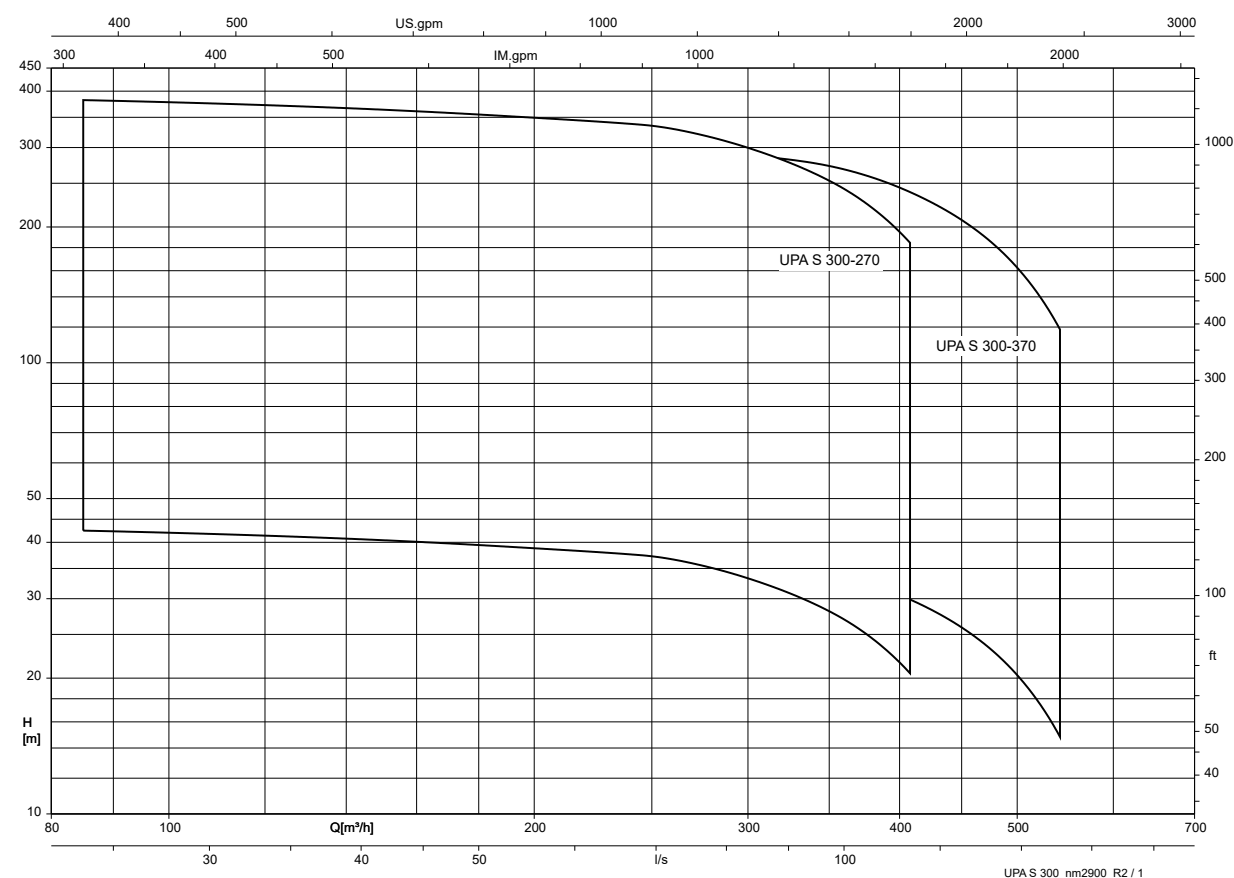
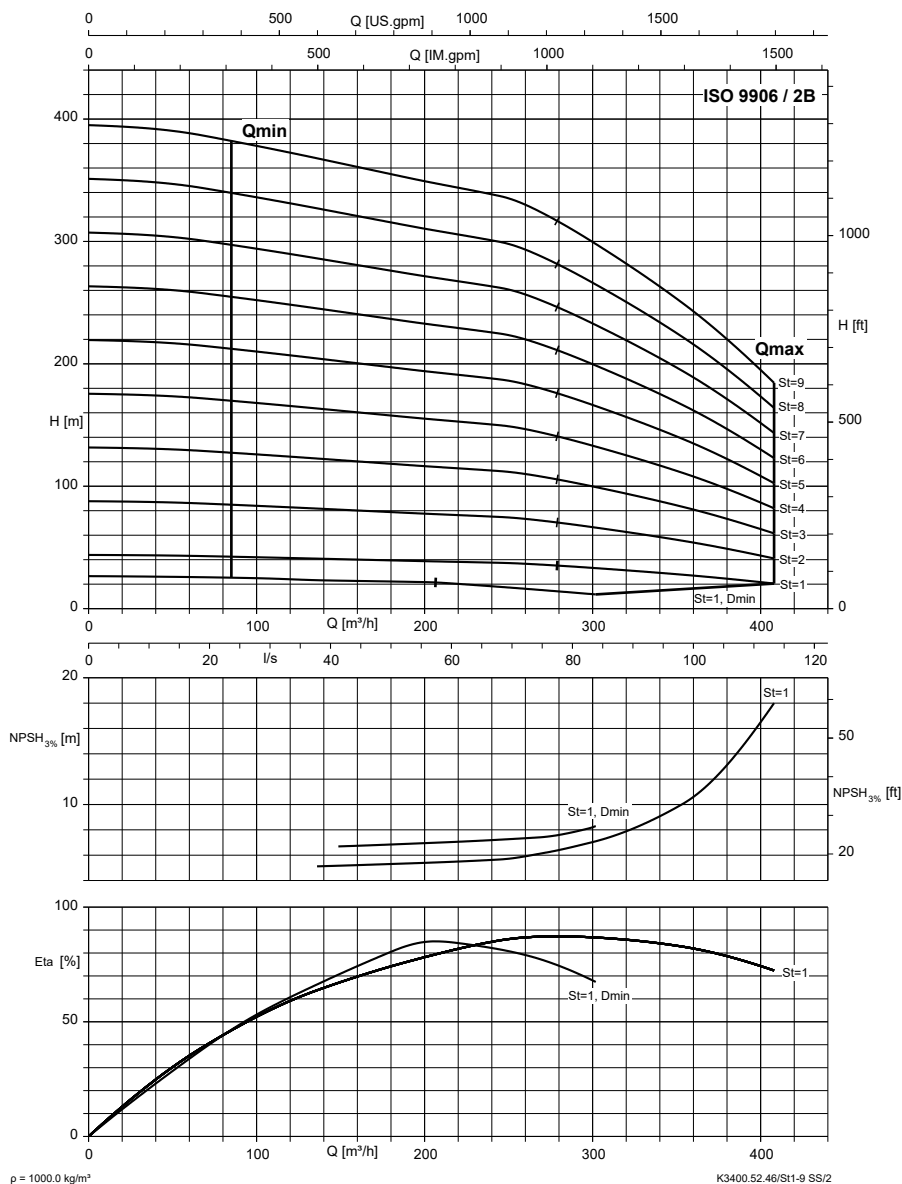


Fig. 1: Selection chart UPA S 300, 50 Hz

UPA S 300 - 270 / ..., number of stages 1 - 9, 50 Hz, n = 2900 rpm

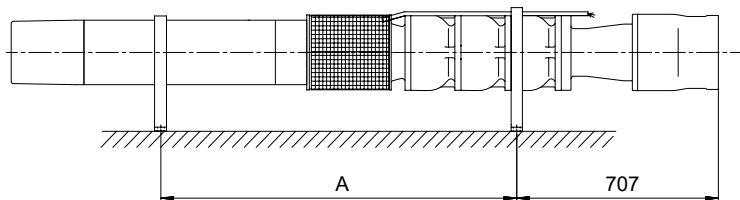


Operating range

$Q_{min} = 85 \text{ m}^3/\text{h}$
 $Q_{max} = \text{end of stage curve}$

The characteristic curves do not take into account the head loss caused by the lift check valve.

1. Refer to the corresponding loss curve to determine and take into account the head loss caused by the lift check valve. ($\Rightarrow$ Fig. 4) ($\Rightarrow$ Fig. 5)



Dimensions of UPA S 300 - 270 / ... [mm]

Connection types

Standard pump end = G8

Alternative pump ends:

- G6
- DN 150, DN 200
- NPT6-8, NPT8-8

i On versions with an alternative pump end, dimensions D_{max} and L_A differ from those with the standard pump end.

For alternative pump end dimensions see: ($\Rightarrow$ Table 26)

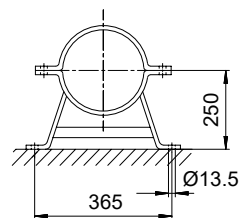
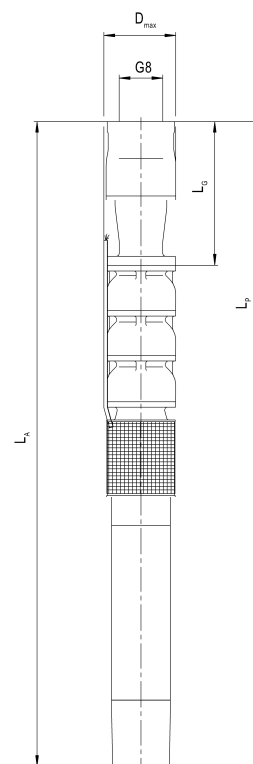


Table 8: Dimensions, weights and installation type depending on the motor, 50 Hz / 400 V 3~

UPA S 300 - 270 / ...	A	L _P ²⁾	L _A ²⁾	L _G ³⁾²⁾	D _{max} ²⁾		Total weight	Type of installation	
					DOL	Y - Δ		Vertical	Horizontal
					[mm]	[mm]	Material variant 1.4408 / 1.4517 [kg]		
1 + UMA 200-37/21	900	1037	2177	535	277	282	261	X	X
2 + UMA 200-75/21	1302	1229	2789	535	282	282	367	X	X
2 + UMA 250-85/21	1262	1259	2678	535	285	285	442	X	X
3 + UMA 250-110/21	1509	1451	2980	535	285	287	509	X	X
4 + UMA 250-160/21	1821	1643	3412	535	287	305	620	X	X
5 + UMA 250-190/21	-	1835	3754	535	305	305	702	X	-
6 + UMA 300-250/22	-	2027	4100	535	310	315	861	X	-
7 + UMA 300-250/22	-	2219	4292	535	310	315	891	X	-
8 + UMA 300-300/22	-	2411	4664	535	315	315	999	X	-
9 + UMA 300-400/22	-	2603	4976	535	315	315	1081	X	-

Table 9: Technical data for 50 Hz / 400 V 3~, n = 2900 rpm, flow velocity past the motor v = 0.2 m/s

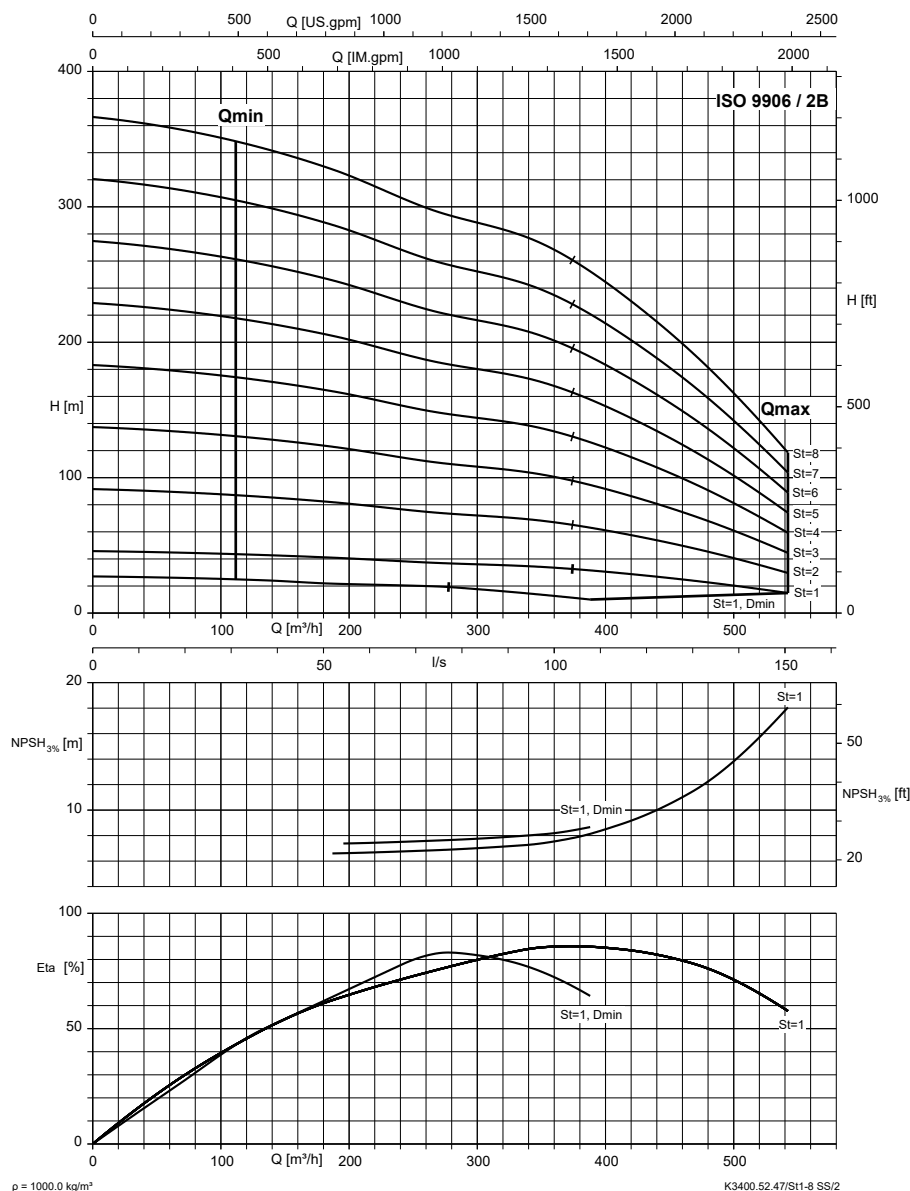
UPA S 300 - 270 / ...	Pump		Motor			Motor lead	
	Head at Q = 0 m³/h	Maximum power in- put	Rated power	Rated current	Effi- ciency	Number × cross-section of cores ⁴⁾	
						DOL	Y - Δ
						[mm²]	[mm²]
1 + UMA 200-37/21	44	32	37	75	85,0	F4 × 10	F3/F4 × 6
2 + UMA 200-75/21	88	65	75	150	87,3	F3/F4 × 10 II	F3/F4 × 10
2 + UMA 250-85/21	88	65	85	163	87,4	F3/F4 × 10 II	F3/F4 × 16
3 + UMA 250-110/21	132	97	110	215	88,5	F3/F4 × 16 II	F3/F4 × 25
4 + UMA 250-160/21	176	129	160	301	89,1	F3/F4 × 25 II	F3/R4 × 35
5 + UMA 250-190/21	219	161	190	365	89,7	F3/R4 × 35 II	F3/R4 × 35
6 + UMA 300-250/22	263	197	250	465	88,4	F3/F3 × 70 II + R1 × 95	6 × R1 × 95 + R1 × 95
7 + UMA 300-250/22	307	226	250	465	88,4	F3/F3 × 70 II + R1 × 95	6 × R1 × 95 + R1 × 95
8 + UMA 300-300/22	351	258	300	558	89,2	6 × R1 × 95 II + R1 × 95	6 × R1 × 95 + R1 × 95
9 + UMA 300-400/22	395	291	400	739	89,6	6 × R1 × 95 II + R1 × 95	6 × R1 × 95 + R1 × 95

Table 10: Explanation of abbreviations

Abbreviation	Explanation
II	Parallel cables
F4	1 × 4-core, cable cross-section: flat
R1	1 × 1-core, cable cross-section: round
F3/F3	1 × 3-core, cable cross-section: flat + 1 × 3-core, cable cross-section: flat
F3/F4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°
F3/R4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°

- 2 Dimension indicated refers to version with lift check valve and standard pump end G8.
3 Length of pump end fitted as standard
4 Designed for submerged use, 400 V, ≤ +30 °C

UPA S 300 - 370 / ..., number of stages 1 - 8, 50 Hz, n = 2900 rpm



Connection types

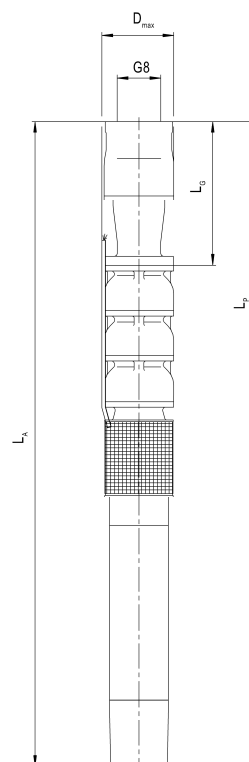
Standard pump end = G8

Alternative pump ends:

- G10
- DN 200, DN 250
- NPT8-8, NPT10-8

i On versions with an alternative pump end, dimensions D_{max} and L_A differ from those with the standard pump end.

For alternative pump end dimensions see: (⇒ Table 27)



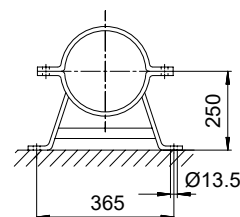
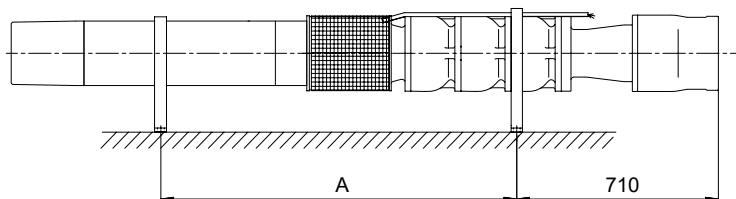
Operating range

$Q_{min} = 112 \text{ m}^3/\text{h}$

$Q_{max} = \text{end of stage curve}$

The characteristic curves do not take into account the head loss caused by the lift check valve.

1. Refer to the corresponding loss curve to determine and take into account the head loss caused by the lift check valve. (⇒ Fig. 7) (⇒ Fig. 6)



Dimensions of UPA S 300 - 370 / ... [mm]

Table 11: Dimensions, weights and installation type depending on the motor, 50 Hz / 400 V 3~

UPA S 300 - 370 / ...	A	L _P ⁵⁾	L _A ⁵⁾	L _G ⁶⁾⁵⁾	D _{max} ⁵⁾		Total weight	Type of installation	
					DOL	Y - Δ		Vertical	Horizontal
					[mm]	[mm]	Material variant 1.4408 / 1.4517 [kg]		
1 + UMA 200-45/21	945,0	1040	2270	535	282	282	274	X	X
2 + UMA 200-90/21	-	1235	2975	535	281	281	395	X	-
2 + UMA 250-85/21	1264,5	1265	2684	535	285	285	438	X	X
3 + UMA 250-132/21	1579,5	1460	3119	535	287	287	548	X	X
4 + UMA 250-190/21	-	1655	3574	535	305	305	666	X	-
5 + UMA 300-250/22	-	1850	3923	535	310	315	824	X	-
6 + UMA 300-250/22	-	2045	4118	535	310	315	853	X	-
7 + UMA 300-300/22	-	2240	4493	535	315	315	960	X	-
8 + UMA 300-400/22	-	2435	4808	535	315	315	1042	X	-

Table 12: Technical data for 50 Hz / 400 V 3~, n = 2900 rpm, flow velocity past the motor v = 0.2 m/s

UPA S 300 - 370 / ...	Pump		Motor			Motor lead	
	Head at Q = 0 m³/h	Maximum power input	Rated power	Rated current	Efficiency	Number × cross-section of cores ⁷⁾	
	H ₀ [m]	P ₂ [kW]	P _N [kW]	I _N [A]	η _M [%]	DOL [mm²]	Y - Δ [mm²]
1 + UMA 200-45/21	46	39	45	90	86,0	F4 × 10 II	F3/F4 × 6
2 + UMA 200-90/21	91	78	90	180	88,0	F3/F4 × 16 II	F3/F4 × 16
2 + UMA 250-85/21	91	78	85	163	87,4	F3/F4 × 10 II	F3/F4 × 16
3 + UMA 250-132/21	137	118	132	249	89,0	F3/F4 × 25 II	F3/F4 × 25
4 + UMA 250-190/21	183	157	190	365	89,7	F3/R4 × 35 II	F3/R4 × 35
5 + UMA 300-250/22	228	196	250	465	88,4	F3/F3 × 70 II + R1 × 95	6 × R1 × 95 + R1 × 95
6 + UMA 300-250/22	274	235	250	465	88,4	F3/F3 × 70 II + R1 × 95	6 × R1 × 95 + R1 × 95
7 + UMA 300-300/22	319	275	300	558	89,2	6 × R1 × 95 II + R1 × 95	6 × R1 × 95 + R1 × 95
8 + UMA 300-400/22	365	314	400	739	89,6	6 × R1 × 95 II + R1 × 95	6 × R1 × 95 + R1 × 95

Table 13: Explanation of abbreviations

Abbreviation	Explanation
II	Parallel cables
F4	1 × 4-core, cable cross-section: flat
R1	1 × 1-core, cable cross-section: round
F3/F3	1 × 3-core, cable cross-section: flat + 1 × 3-core, cable cross-section: flat
F3/F4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°
F3/R4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°

⁵⁾ Dimension indicated refers to version with lift check valve and standard pump end G8.

⁶⁾ Length of pump end fitted as standard

⁷⁾ Designed for submerged use, 400 V, ≤ +30 °C

Technical data, 60 Hz

Tolerance to ISO 9906 Cl. 2B

The characteristic curves shown are valid for:

- Water at 20 °C
- Density 998 kg/m³
- Viscosity 1.00 mm²/s

Information on characteristic curves

The characteristic curves shown are intended to allow pre-selection. Refer to the quotation for the precise selection data.

Selection chart UPA S 300, 60 Hz, n = 3500 rpm

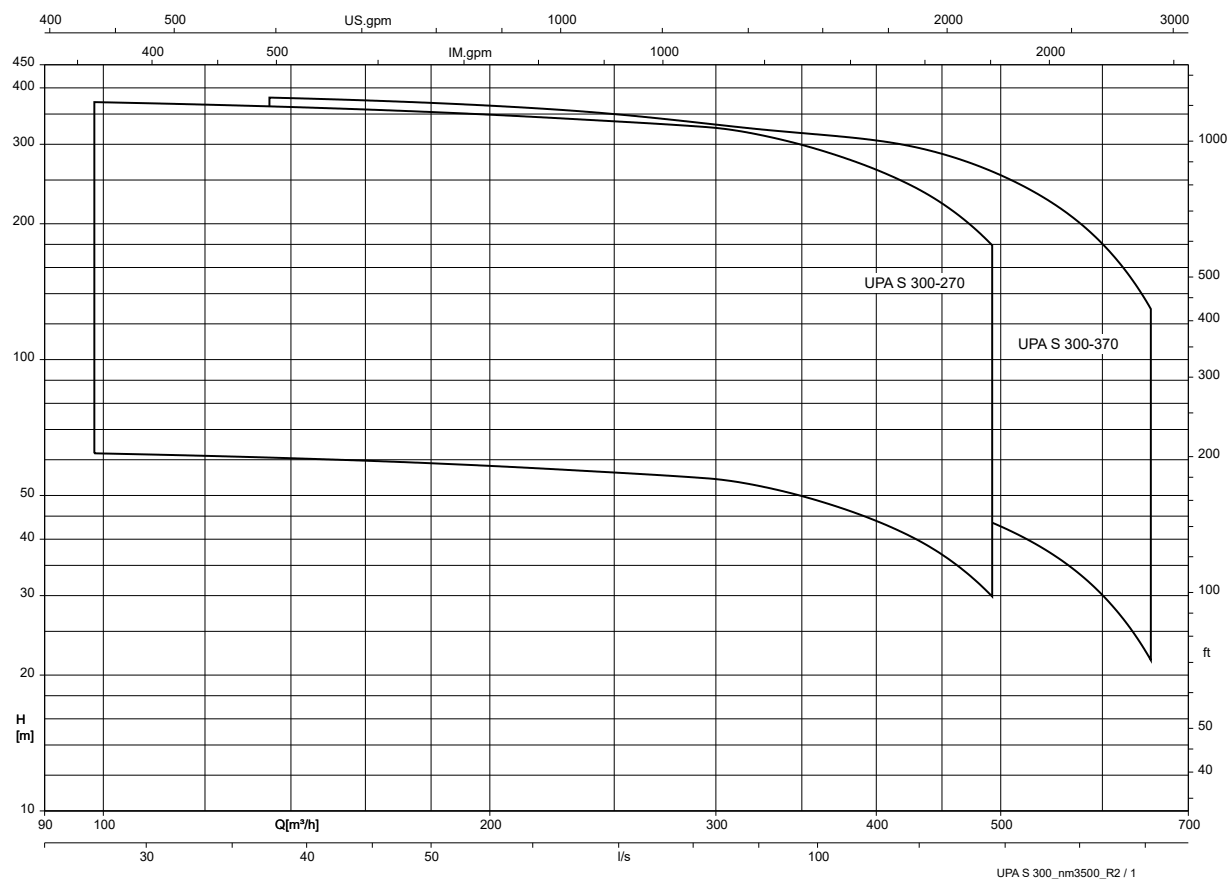
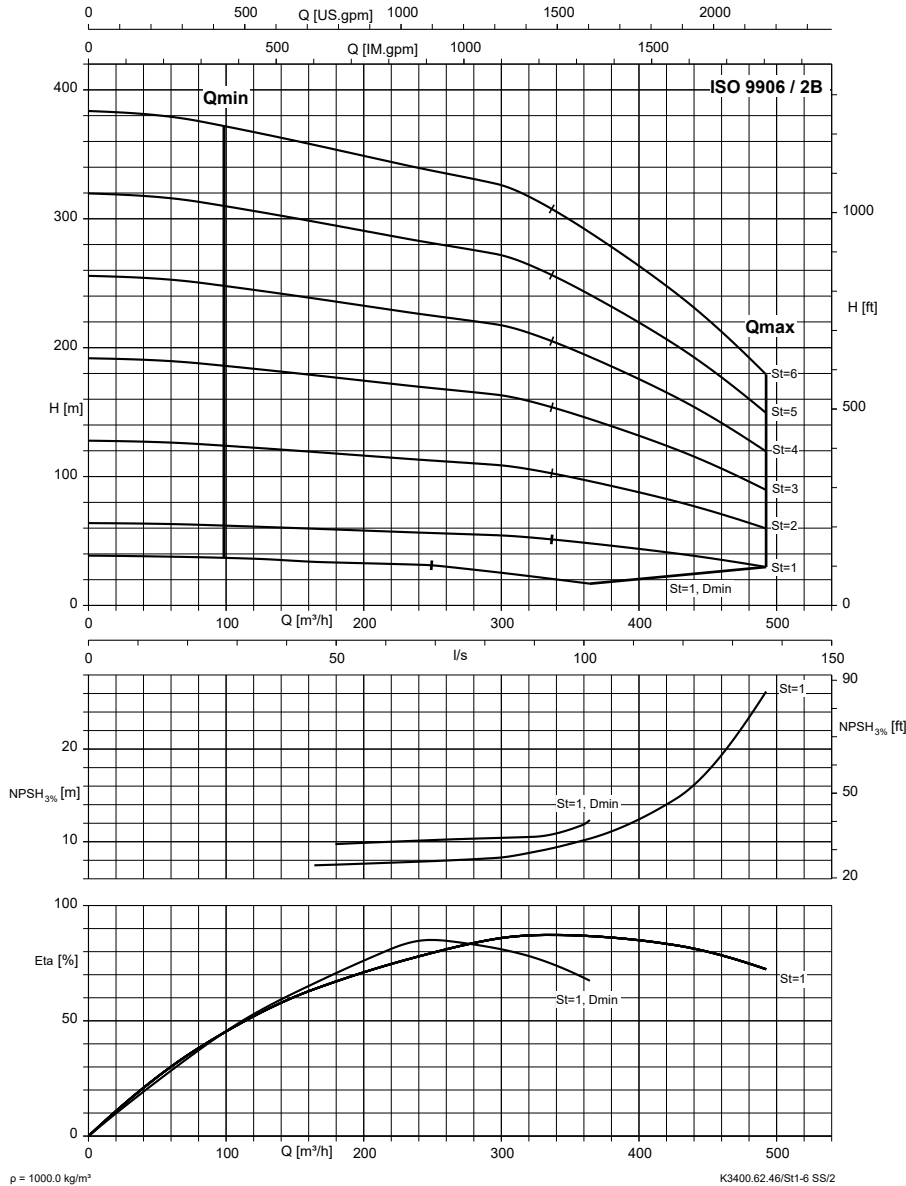


Fig. 2: Selection chart UPA S 300, 60 Hz

UPA S 300 - 270 / ..., number of stages 1 - 6, 60 Hz, n = 3500 rpm



Connection types

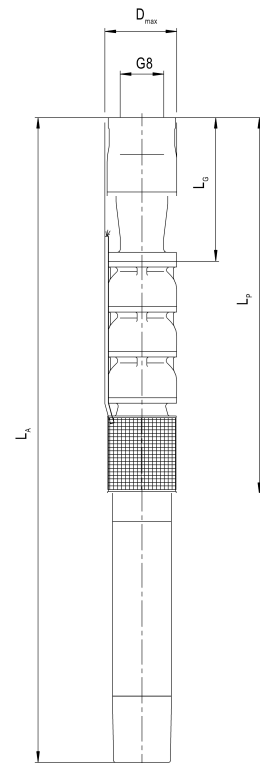
Standard pump end = G8

Alternative pump ends:

- G6
- DN 150, DN 200
- NPT6-8, NPT8-8

i On versions with an alternative pump end, dimensions D_{max} and L_A differ from those with the standard pump end.

For alternative pump end dimensions see: (⇒ Table 26)



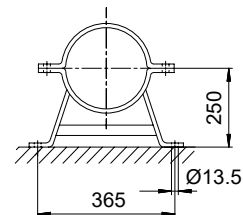
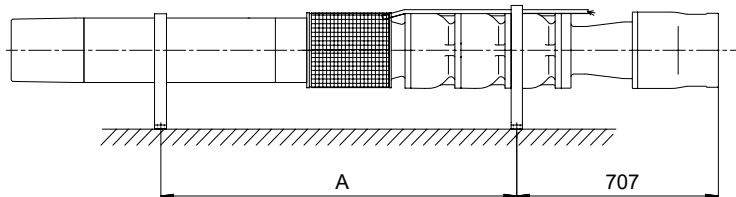
Operating range

$Q_{\text{min}} = 98 \text{ m}^3/\text{h}$

Q_{max} = end of stage curve

The characteristic curves do not take into account the head loss caused by the lift check valve.

1. Refer to the corresponding loss curve to determine and take into account the head loss caused by the lift check valve. (⇒ Fig. 4) (⇒ Fig. 5)



Dimensions of UPA S 300 - 270 / ... [mm]

Table 14: Dimensions, weights and installation type depending on the motor, 60 Hz / 460 V 3~

UPA S 300 - 270 / ...	A	L _P ⁸⁾	L _A ⁸⁾	L _G ⁹⁾⁸⁾	D _{max} ⁸⁾		Total weight	Type of installation	
					DOL	Y - Δ		Vertical	Horizontal
					[mm]	[mm]	Material variant 1.4408 / 1.4517 [kg]		
1 + UMA 200-55/21	1000	1037	2377	535	277	282	297	X	X
2 + UMA 250-132/21	1383	1259	2918	535	285	285	523	X	X
3 + UMA 250-160/21	1630	1451	3220	535	287	305	590	X	X
4 + UMA 300-250/22	-	1643	3716	535	310	310	800	X	-
5 + UMA 300-250/22	-	1835	3908	535	310	310	831	X	-
6 + UMA 300-300/22	-	2027	4280	535	315	315	939	X	-

Table 15: Technical data for 60 Hz / 460 V 3~, n = 3500 rpm, flow velocity past the motor v = 0.2 m/s

UPA S 300 - 270 / ...	Pump		Motor			Motor lead	
	Head at Q = 0 m³/h	Maximum power in- put	Rated power	Rated current	Effi- ciency	Number × cross-section of cores ¹⁰⁾	
	H ₀	P ₂	P _N	I _N	η _M	DOL	Y - Δ
	[m]	[kW]	[kW]	[A]	[%]	[mm²]	[mm²]
1 + UMA 200-55/21	64	57	66	110	86,7	F4 × 16	F3/F4 × 6
2 + UMA 250-132/21	128	114	158	256	89,2	F3/F4 × 25 II	F3/F4 × 25
3 + UMA 250-160/21	192	170	192	310	89,3	F3/F4 × 25 II	F3/R4 × 35
4 + UMA 300-250/22	256	227	300	487	88,5	F3/F3 × 70 II + R1 × 95	F3/F3 × 70 + R1 × 95
5 + UMA 300-250/22	320	284	300	487	88,5	F3/F3 × 70 II + R1 × 95	F3/F3 × 70 + R1 × 95
6 + UMA 300-300/22	384	341	360	585	89,3	6 × R1 × 95 II + R1 × 95	6 × R1 × 95 + R1 × 95

Table 16: Explanation of abbreviations

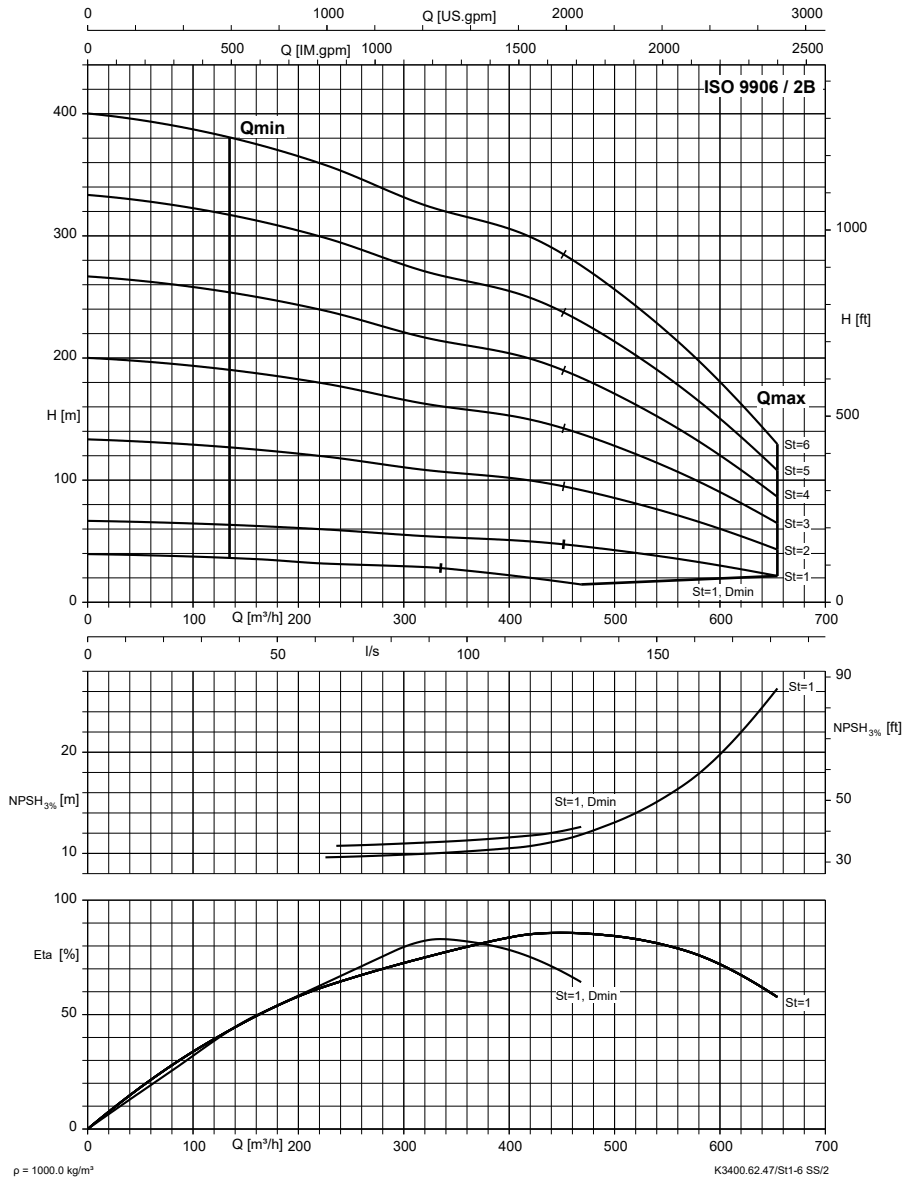
Abbreviation	Explanation
II	Parallel cables
F4	1 × 4-core, cable cross-section: flat
R1	1 × 1-core, cable cross-section: round
F3/F3	1 × 3-core, cable cross-section: flat + 1 × 3-core, cable cross-section: flat
F3/F4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°
F3/R4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°

⁸ Dimension indicated refers to version with lift check valve and standard pump end G8.

⁹ Length of pump end fitted as standard

¹⁰ Designed for submerged use, 460 V, ≤ +30 °C

UPA S 300 - 370 / ..., number of stages 1 - 6, 60 Hz, n = 3500 rpm



Connection types

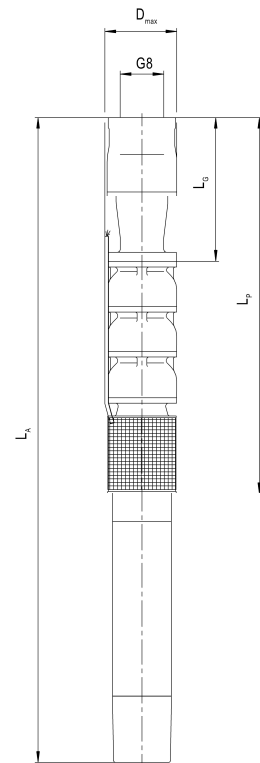
Standard pump end = G8

Alternative pump ends:

- G10
- DN 200, DN 250
- NPT8-8, NPT10-8

i On versions with an alternative pump end, dimensions D_{max} and L_A differ from those with the standard pump end.

For alternative pump end dimensions see: (⇒ Table 27)



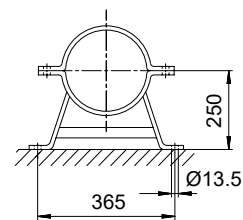
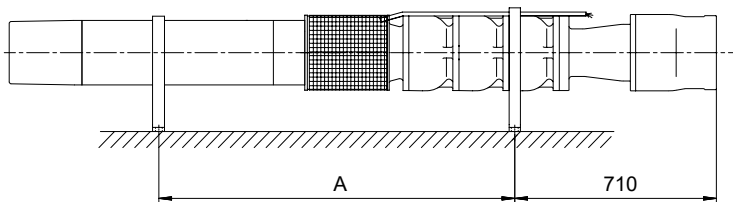
Operating range

$Q_{min} = 135 \text{ m}^3/\text{h}$

Q_{max} = end of stage curve

The characteristic curves do not take into account the head loss caused by the lift check valve.

1. Refer to the corresponding loss curve to determine and take into account the head loss caused by the lift check valve. (⇒ Fig. 7) (⇒ Fig. 6)



Dimensions of UPA S 300 - 370 / ... [mm]

Table 17: Dimensions, weights and installation type depending on the motor, 60 Hz / 460 V 3~

UPA S 300 - 370 / ...	A	L _p ¹¹⁾	L _A ¹¹⁾	L _G ¹²⁾¹¹⁾	D _{max} ¹¹⁾		Total weight	Type of installation	
					DOL	Y - Δ		Vertical	Horizontal
					[mm]	[mm]	Material variant 1.4408 / 1.4517 [kg]		
1 + UMA 200-65/21	1065	1040	2510	535	282	282	317	X	X
2 + UMA 250-132/21	1385	1265	2924	535	287	287	519	X	X
3 + UMA 250-190/21	-	1460	3379	535	305	305	636	X	-
4 + UMA 300-250/22	-	1655	3728	535	310	310	794	X	-
5 + UMA 300-300/22	-	1850	4103	535	315	315	902	X	-
6 + UMA 300-400/22	-	2045	4418	535	315	315	983	X	-

Table 18: Technical data for 60 Hz / 460 V 3~, n = 3500 rpm, flow velocity past the motor v = 0.2 m/s

UPA S 300 - 370 / ...	Pump		Motor			Motor lead	
	Head at Q = 0 m³/h	Maximum power in- put	Rated power	Rated current	Effi- ciency	Number × cross-section of cores ¹³⁾	
	H ₀	P ₂	P _N	I _N	η _M	DOL	Y - Δ
	[m]	[kW]	[kW]	[A]	[%]	[mm²]	[mm²]
1 + UMA 200-65/21	66	69	77	128	87,6	F3/F4 × 6 II	F3/F4 × 10
2 + UMA 250-132/21	133	138	158	256	89,2	F3/F4 × 25 II	F3/F4 × 25
3 + UMA 250-190/21	199	207	228	371	90,0	F3/R4 × 35 II	F3/R4 × 50
4 + UMA 300-250/22	266	276	300	487	88,5	F3/F3 × 70 II + R1 × 95	F3/F3 × 70 + R1 × 95
5 + UMA 300-300/22	332	345	360	585	89,3	6 × R1 × 95 II + R1 × 95	6 × R1 × 95 + R1 × 95
6 + UMA 300-400/22	399	414	420	682	89,9	6 × R1 × 95 II + R1 × 95	6 × R1 × 95 + R1 × 95

Table 19: Explanation of abbreviations

Abbreviation	Explanation
II	Parallel cables
R1	1 × 1-core, cable cross-section: round
F3/F3	1 × 3-core, cable cross-section: flat + 1 × 3-core, cable cross-section: flat
F3/F4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°
F3/R4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°

11 Dimension indicated refers to version with lift check valve and standard pump end G8.

12 Length of pump end fitted as standard

13 Designed for submerged use, 460 V, ≤ +30 °C

Technical data, 100 Hz

Tolerance to ISO 9906 Cl. 2B

The characteristic curves shown are valid for:

- Water at 20 °C
- Density 998 kg/m³
- Viscosity 1.00 mm²/s

Information on characteristic curves

The characteristic curves shown refer to pump sets driven by a UMA-S synchronous motor, n = 3000 rpm.

The characteristic curves shown are intended to allow pre-selection. Refer to the quotation for the precise selection data.

Selection chart UPA S 300, 100 Hz, n = 3000 rpm

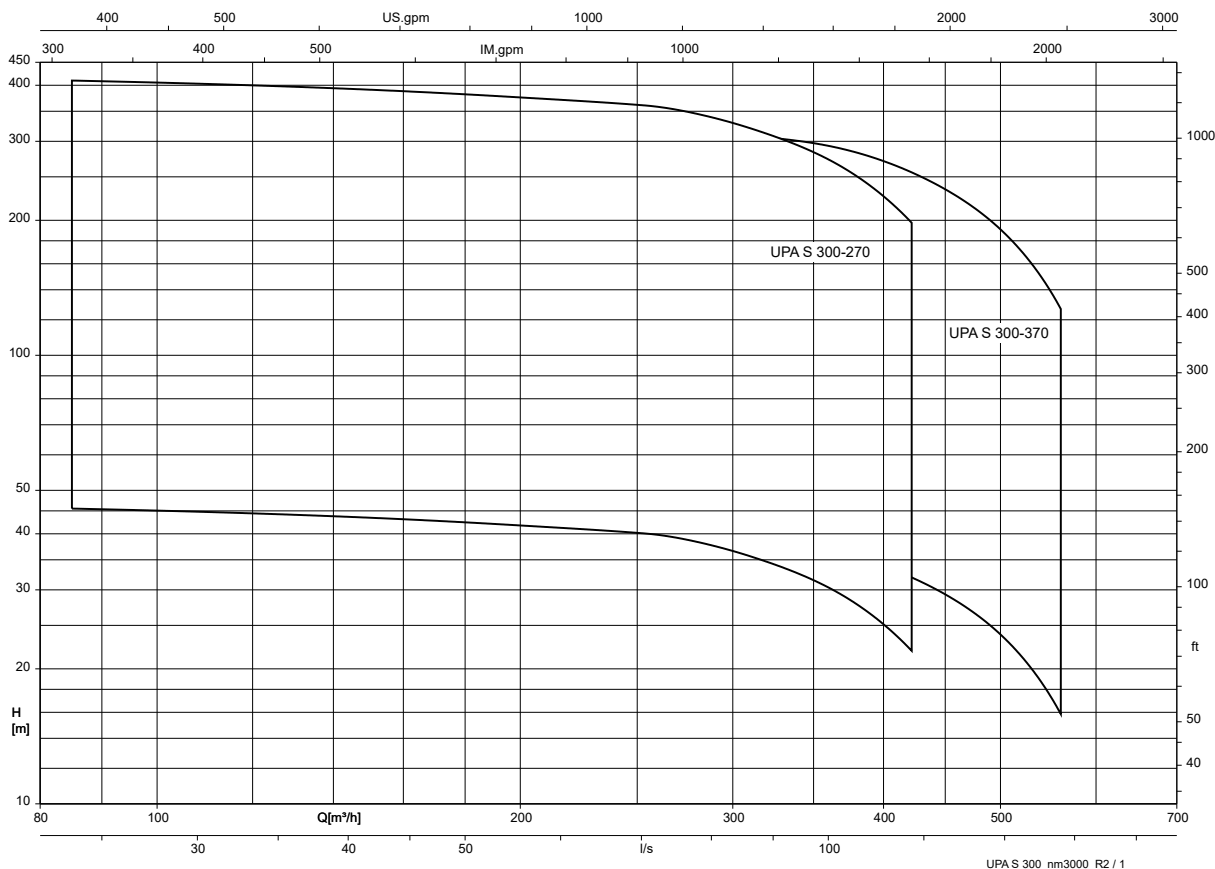
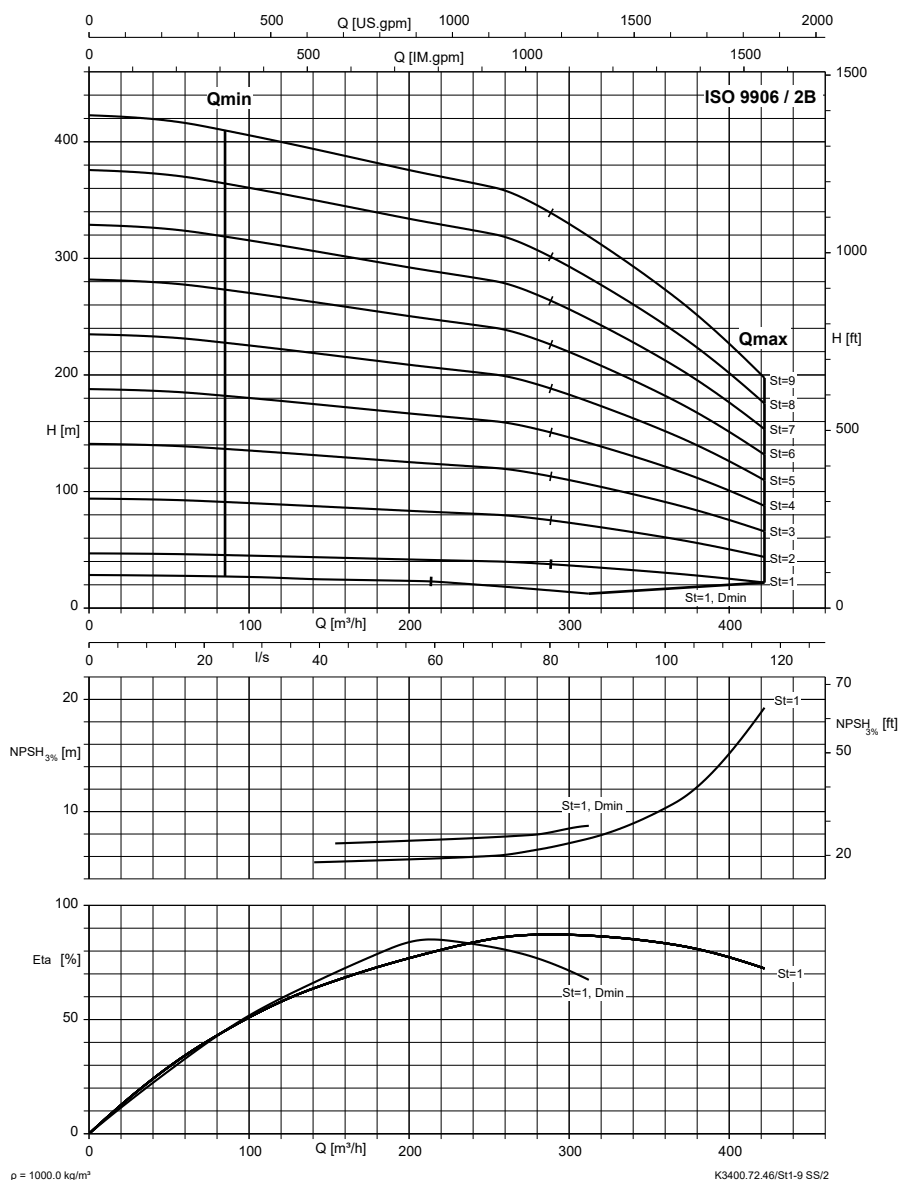


Fig. 3: Selection chart UPA S 300, 100 Hz

UPA S 300 - 270 / ..., number of stages 1 - 9, 100 Hz, n = 3000 rpm



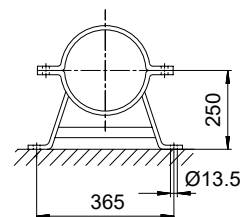
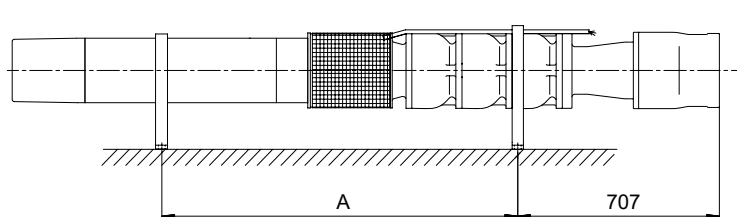
Operating range

$$Q_{\min} = 85 \text{ m}^3/\text{h}$$

$Q_{\min}$ = end of stage curve

The characteristic curves do not take into account the head loss caused by the lift check valve.

1. Refer to the corresponding loss curve to determine and take into account the head loss caused by the lift check valve. (⇒ Fig. 4) (⇒ Fig. 5)



Dimensions of UPA S 300 - 270 / ... [mm]

Connection types

Standard pump end = G8

Alternative pump ends:

- G6
- DN 150, DN 200
- NPT6-8, NPT8-8

 On versions with an alternative pump end, dimensions $D_{\max}$ and L_A differ from those with the standard pump end.

For alternative pump end dimensions see:
(⇒ Table 26)

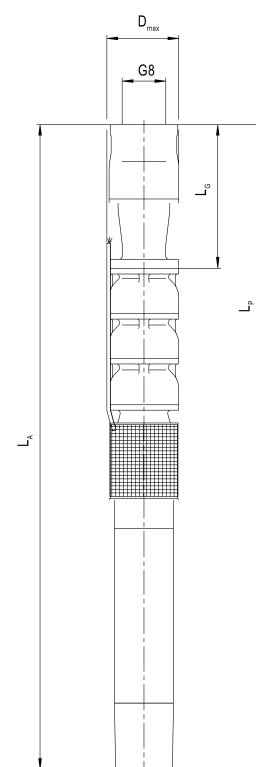


Table 20: Dimensions, weights and installation type depending on the motor, 100 Hz / 400 V 3~

UPA S 300 - 270 / ...	A	L _p ¹⁴⁾	L _A ¹⁴⁾	L _G ¹⁵⁾¹⁴⁾	D _{max} ¹⁴⁾		Total weight	Type of installation	
					VFD	VFD parallel	Material variant 1.4408 / 1.4517	Vertical	Horizontal
	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[kg]		
1 + UMA-S 200-75/42	933	1037	2242	535	277	281	271	X	X
2 + UMA-S 200-100/42	1180	1229	2545	535	277	281	321	X	X
3 + UMA-S 200-130/42	1456	1421	2905	535	-	281	379	X	X
4 + UMA-S 250-200/42	1766	1643	3302	535	-	305	521	X	X
5 + UMA-S 250-200/42	-	1835	3494	535	-	305	552	X	-
6 + UMA-S 250-250/42	-	2027	3796	535	-	301	633	X	-

Table 21: Technical data for 100 Hz / 400 V 3~, n = 3000 rpm, flow velocity past the motor v = 0.2 m/s

UPA S 300 - 270 / ...	Pump		Motor			Motor lead	
	Head at Q = 0 m³/h	Maximum power in- put	Rated power	Rated current	Effi- ciency	Number × cross-section of cores ¹⁶⁾	
	H ₀	P ₂	P _N	I _N	η _M	VFD	VFD parallel
	[m]	[kW]	[kW]	[A]	[%]	[mm²]	[mm²]
1 + UMA-S 200-75/42	47	36	75	128	92,8	F4 × 16	F3/F4 × 16 II
2 + UMA-S 200-100/42	94	71	100	177	92,9	R4 × 25	F3/F4 × 16 II
3 + UMA-S 200-130/42	141	107	130	224	93,3	-	F3/F4 × 16 II
4 + UMA-S 250-200/42	188	143	200	389	93,8	-	F3/R4 × 35 II
5 + UMA-S 250-200/42	235	179	200	389	93,8	-	F3/R4 × 35 II
6 + UMA-S 250-250/42	282	214	250	493	93,8	-	6 × R1 × 35 II + R1 × 35

Table 22: Explanation of abbreviations

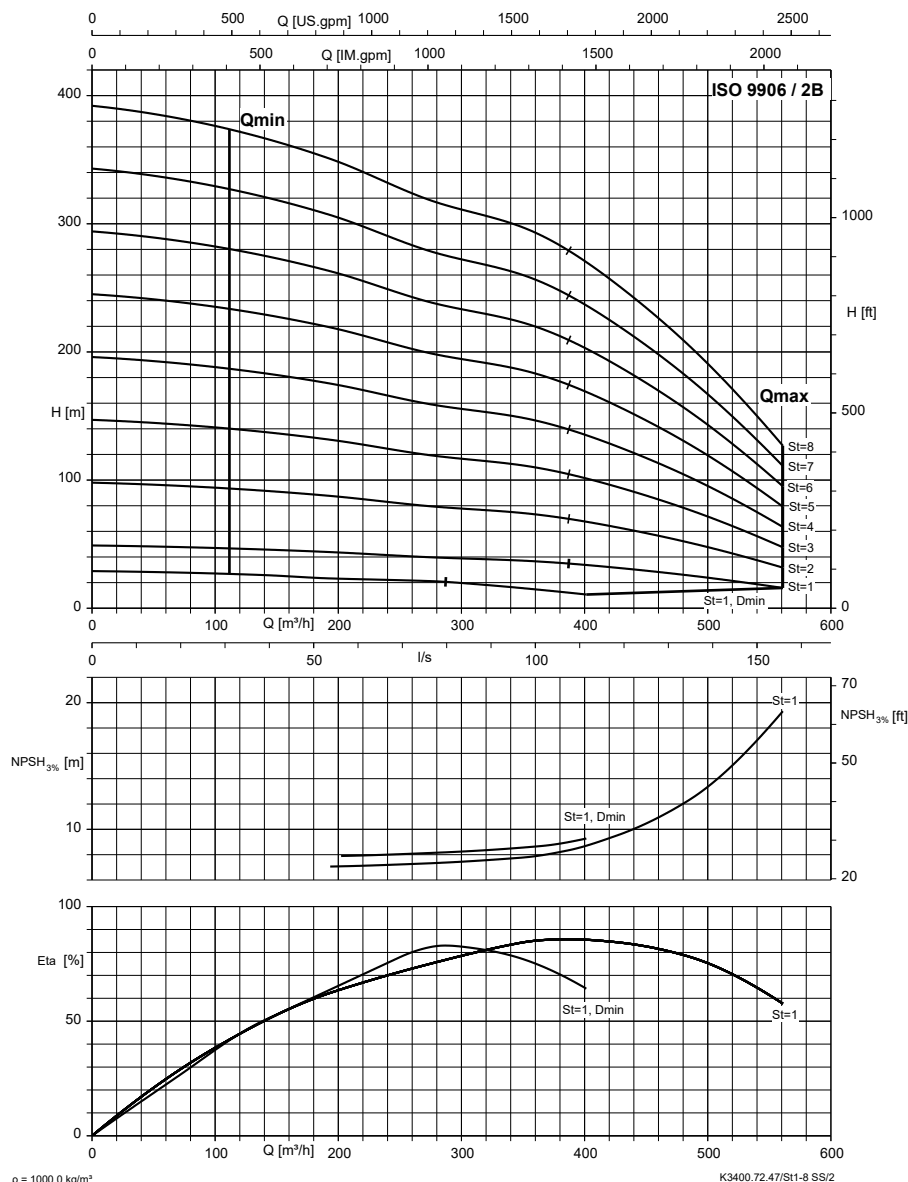
Abbreviation	Explanation
II	Parallel cables
F4	1 × 4-core, cable cross-section: flat
R1	1 × 1-core, cable cross-section: round
R4	1 × 4-core, cable cross-section: round
F3/F4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°
F3/R4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°

¹⁴ Dimension indicated refers to version with lift check valve and standard pump end G8.

¹⁵ Length of pump end fitted as standard

¹⁶ Designed for submerged use, 400 V, ≤ +30 °C

UPA S 300 - 370 / ..., number of stages 1 - 8, 100 Hz, n = 3000 rpm



Connection types

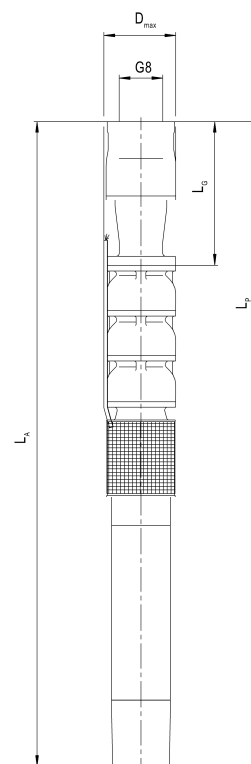
Standard pump end = G8

Alternative pump ends:

- G10
- DN 200, DN 250
- NPT8-8, NPT10-8

i On versions with an alternative pump end, dimensions D_{max} and L_A differ from those with the standard pump end.

For alternative pump end dimensions see: (⇒ Table 27)



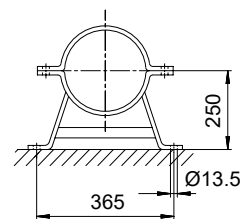
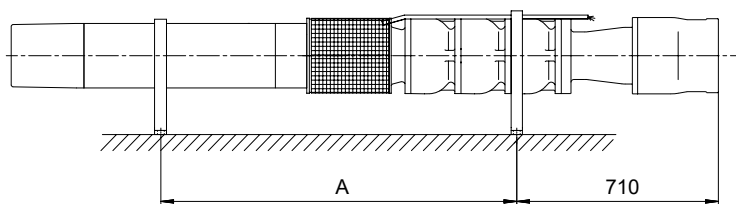
Operating range

$Q_{min} = 112 \text{ m}^3/\text{h}$

Q_{max} = end of stage curve

The characteristic curves do not take into account the head loss caused by the lift check valve.

1. Refer to the corresponding loss curve to determine and take into account the head loss caused by the lift check valve. (⇒ Fig. 7) (⇒ Fig. 6)



Dimensions of UPA S 300 - 370 / ... [mm]

Table 23: Dimensions, weights and installation type depending on the motor, 100 Hz / 400 V 3~

UPA S 300 - 370 / ...	A	L _p ¹⁷⁾	L _A ¹⁷⁾	L _G ¹⁸⁾¹⁷⁾	D _{max} ¹⁷⁾		Total weight	Type of installation	
					VFD	VFD parallel	Material variant 1.4408 / 1.4517	Vertical	Horizontal
	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[kg]		
1 + UMA-S 200-75/42	933	1040	2245	535	277	281	268	X	X
2 + UMA-S 200-100/42	1183	1235	2551	535	296	281	317	X	X
3 + UMA-S 250-200/42	1580	1460	3119	535	-	281	486	X	X
4 + UMA-S 250-200/42	1775	1655	3314	535	-	305	516	X	X
5 + UMA-S 250-250/42	-	1850	3619	535	-	301	596	X	-

Table 24: Technical data for 100 Hz / 400 V 3~, n = 3000 rpm, flow velocity past the motor v = 0.2 m/s

UPA S 300 - 370 / ...	Pump		Motor			Motor lead	
	Head at Q = 0 m³/h	Maximum power input	Rated power	Rated current	Efficiency	Number × cross-section of cores ¹⁹⁾	
	H ₀	P ₂	P _N	I _N	η _M	VFD	VFD parallel
	[m]	[kW]	[kW]	[A]	[%]	[mm²]	[mm²]
1 + UMA-S 200-75/42	49	43	75	128	92,8	F4 × 16	F3/F4 × 16 II
2 + UMA-S 200-100/42	98	86	100	177	92,9	R4 × 25	F3/F4 × 16 II
3 + UMA-S 250-200/42	146	129	200	389	93,8	-	F3/R4 × 35 II
4 + UMA-S 250-200/42	195	172	200	389	93,8	-	F3/R4 × 35 II
5 + UMA-S 250-250/42	244	215	250	493	93,8	-	6 × R1 × 35 II + R1 × 35

Table 25: Explanation of abbreviations

Abbreviation	Explanation
II	Parallel cables
F4	1 × 4-core, cable cross-section: flat
R1	1 × 1-core, cable cross-section: round
R4	1 × 4-core, cable cross-section: round
F3/F4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°
F3/R4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°

¹⁷ Dimension indicated refers to version with lift check valve and standard pump end G8.

¹⁸ Length of pump end fitted as standard

¹⁹ Designed for submerged use, 400 V, ≤ +30 °C

Loss curves for lift check valves

Head losses caused by lift check valve for UPA S 300-270, pump end G6/NPT 6-8/DN 150

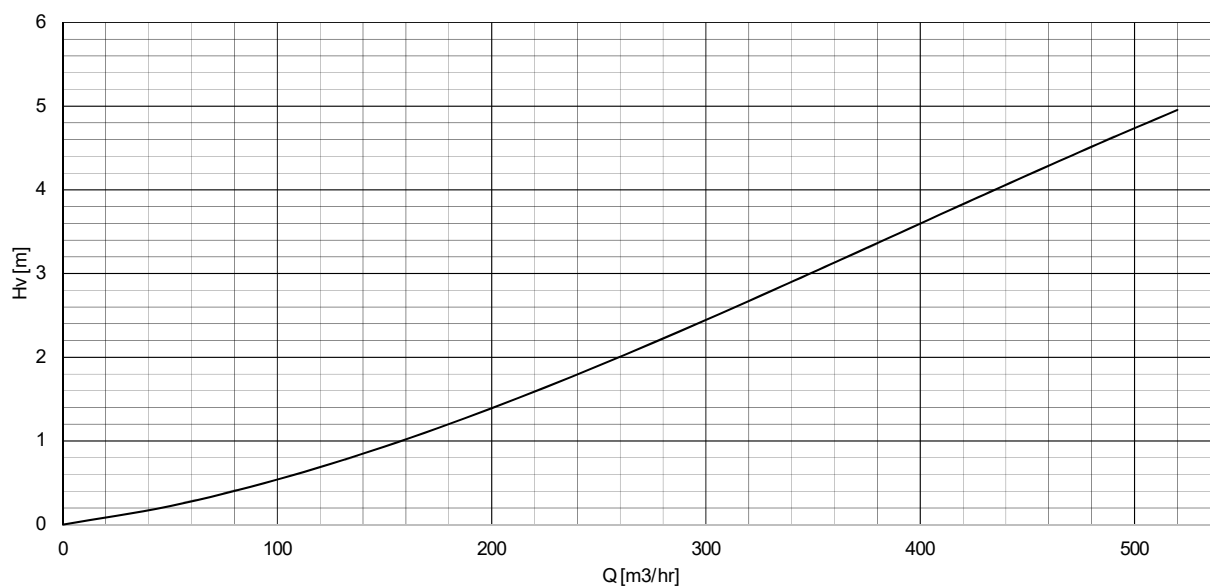


Fig. 4: Loss curve, UPA S 300-270, pump end G6

Head losses caused by lift check valve for UPA S 300-270, pump end G8/NPT 8-8/DN 200

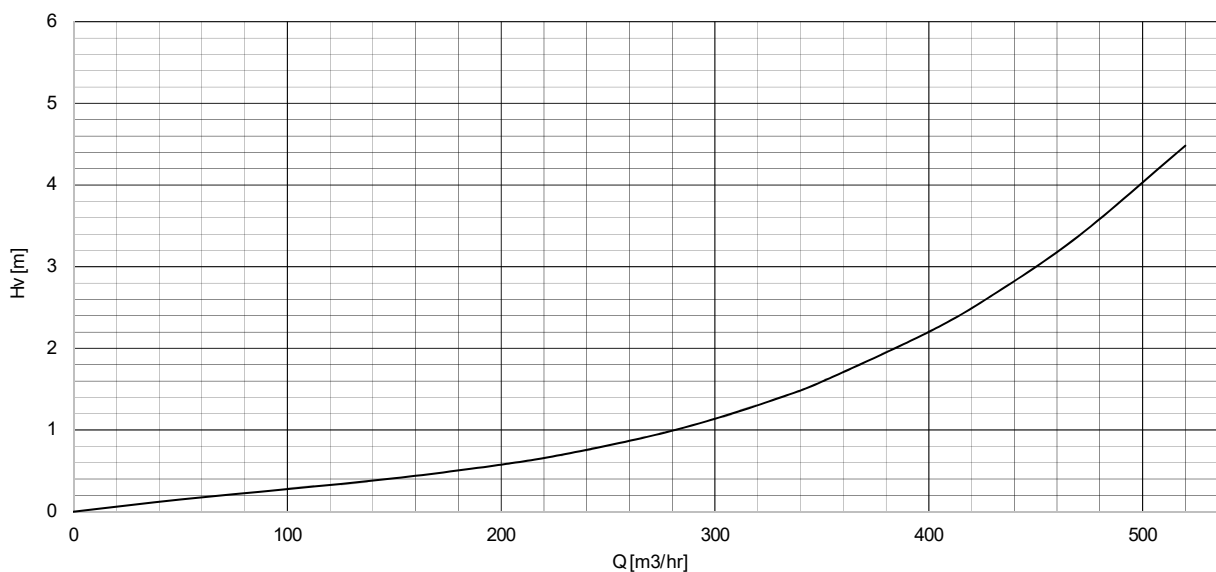


Fig. 5: Loss curve, UPA S 300-270, pump end G8

Head losses caused by lift check valve for UPA S 300-370, pump end G8/NPT 8-8/DN 200

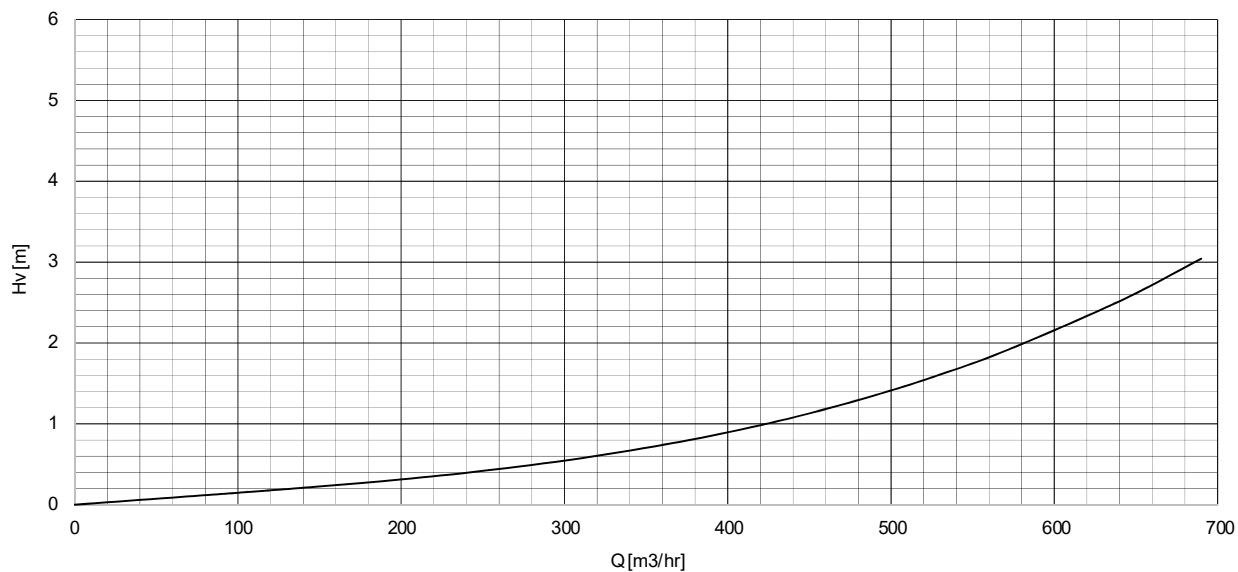


Fig. 6: Loss curve, UPA S 300-370, pump end G8

Head losses caused by lift check valve for UPA S 300-370, pump end G10/NPT 10-8/DN 250

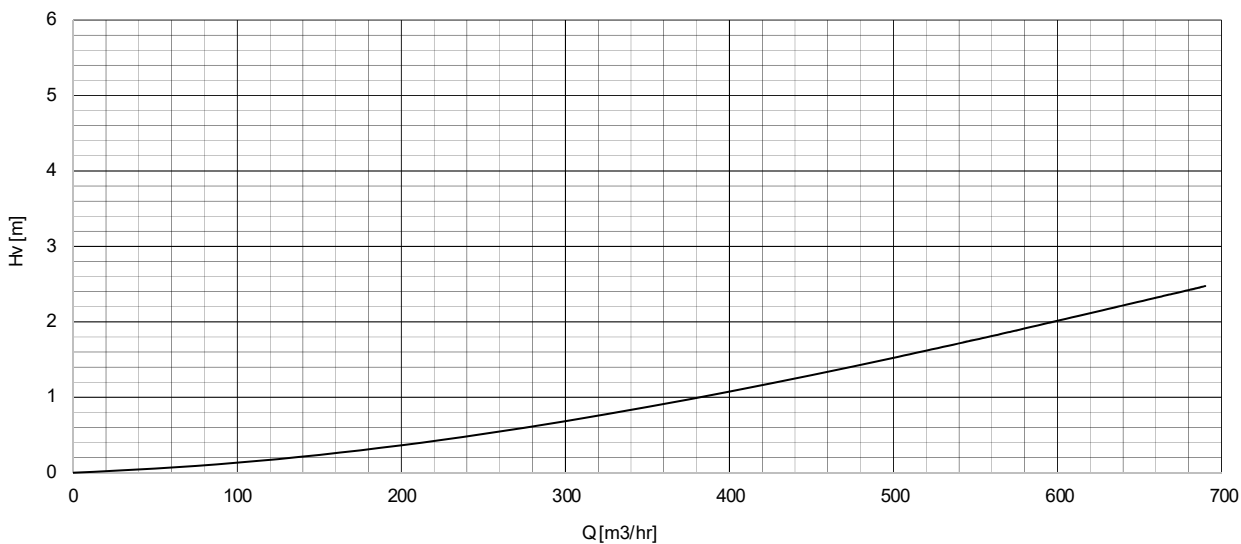


Fig. 7: Loss curve, UPA S 300-370, pump end G10

Dimensions of alternative pump end

i On versions with an alternative pump end, dimensions $D_{\max}^*$ and L_A^* differ from those with the standard pump end.

- The new dimension $D_{\max}^*$ for the selected size can be found in the following tables.
- The new dimension L_A^* must be calculated: $L_A^* = L_A - L_G + L_G^*$
 - Dimensions L_A und L_G for the selected size can be found in the standard "Dimensions" table.
 - Dimension L_G^* for the selected size can be found in the following tables.

Table 26: Dimensions, alternative pump ends, UPA S 300 - 270

Pump end		$D_{\max}^*$	L_G^*
		[mm]	[mm]
Lift check valve / Connection nozzle	G6 / NPT 6-8	315	330
	DN 150 / PN 10, PN 16	315	299
	DN 150 / PN 25, PN 40	320	299
	DN 150 / PN 63	345	299
Lift check valve	G8 ²⁰⁾	315	535
	NPT 8-8	315	535
	DN 200 / PN 10, PN 16	352	495
	DN 200 / PN 25	362	495
	DN 200 / PN 40	375	495
	DN 200 / PN 63	430	495
Connection nozzles	G8 ²⁰⁾	315	460
	NPT 8-8	315	460
	DN 200 / PN 10, PN 16	352	405
	DN 200 / PN 25	362	405
	DN 200 / PN 40	375	405
	DN 200 / PN 63	430	405

Table 27: Dimensions, alternative pump ends, UPA S 300 - 370

Pump end		$D_{\max}^*$	L_G^*
		[mm]	[mm]
Lift check valve	G8 ²⁰⁾	315	535
	NPT 8-8	315	535
	DN 200 / PN 10, PN 16	352	495
	DN 200 / PN 25	362	495
	DN 200 / PN 40	375	495
	DN 200 / PN 63	430	495
	G10 / NPT 10-8	416	720
	DN 250 / PN 10	416	670
	DN 250 / PN 16	416	670
	DN 250 / PN 25	425	670
	DN 250 / PN 40	450	670
	DN 250 / PN 63	470	670
Connection nozzles	G8 ²⁰⁾	315	460
	NPT 8-8	315	460
	DN 200 / PN 10, PN 16	352	405
	DN 200 / PN 25	362	405
	DN 200 / PN 40	375	405
	DN 200 / PN 63	430	405
	G10 / NPT 10-8	416	460
	DN 250 / PN 10	416	460
	DN 250 / PN 16	416	460
	DN 250 / PN 25	425	460
	DN 250 / PN 40	450	460
	DN 250 / PN 63	470	460

Scope of supply

Depending on the model, the following items are included in the scope of supply:

- Pump set with motor lead
- Back-up name plate

Optional:

- Extension cable connected or supplied but not fitted
- Cable connector
- Cable clips
- Pedestals

- Cooling shroud, suction shroud or pressure shroud
- Supporting clamps and mounting clamps
- Electrical protection equipment
- Automatic control units
- Centring device
- Spare parts
- Hydraulic performance test
- Frequency inverter
- Voltage filter
- Dry running protection device

Cross sectional drawings with list of components

UPA S 300-270/370, standard design

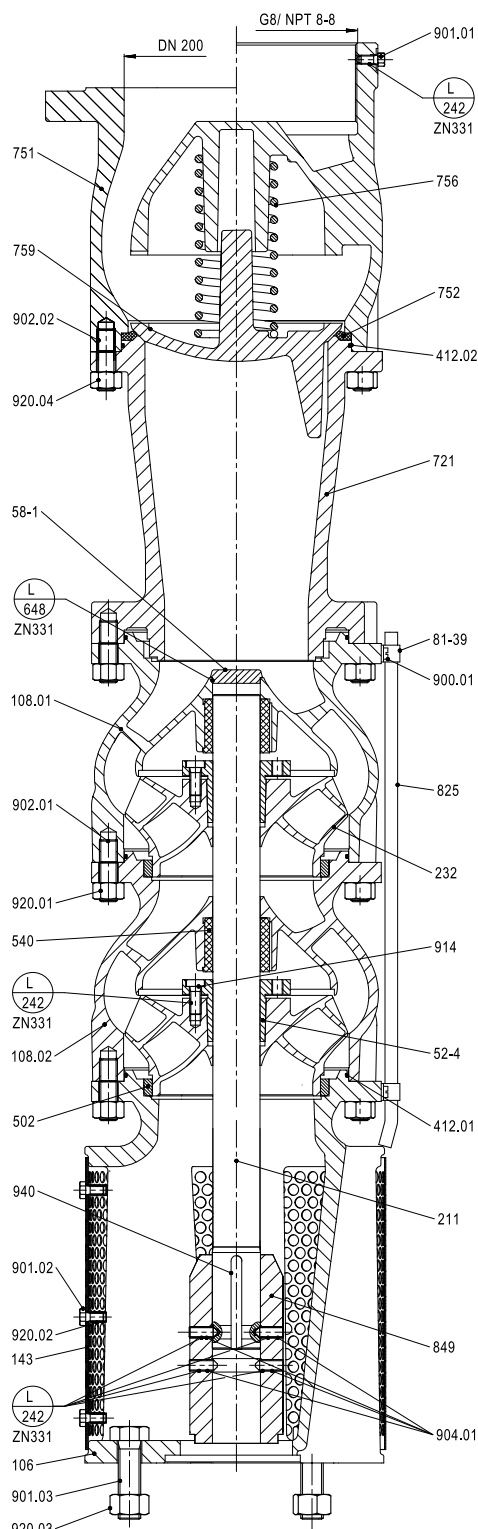


Fig. 8: UPA S 300-270/370 incl. lift check valve G8/NPT 8-8/DN 200, for connection to UMA 250 or UMA-S 250 motor, standard design

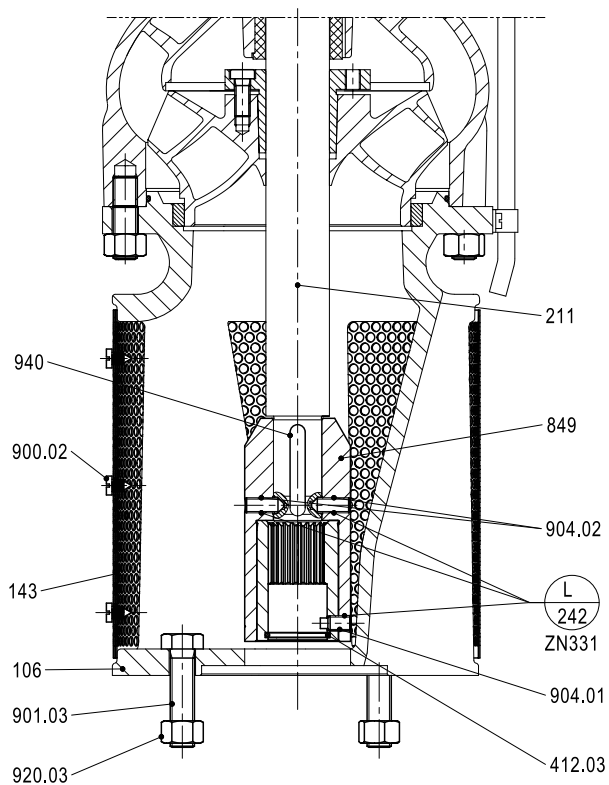


Fig. 9: UPA S 300-270/370 with connection for UMA 200 or UMA-S 200 motor

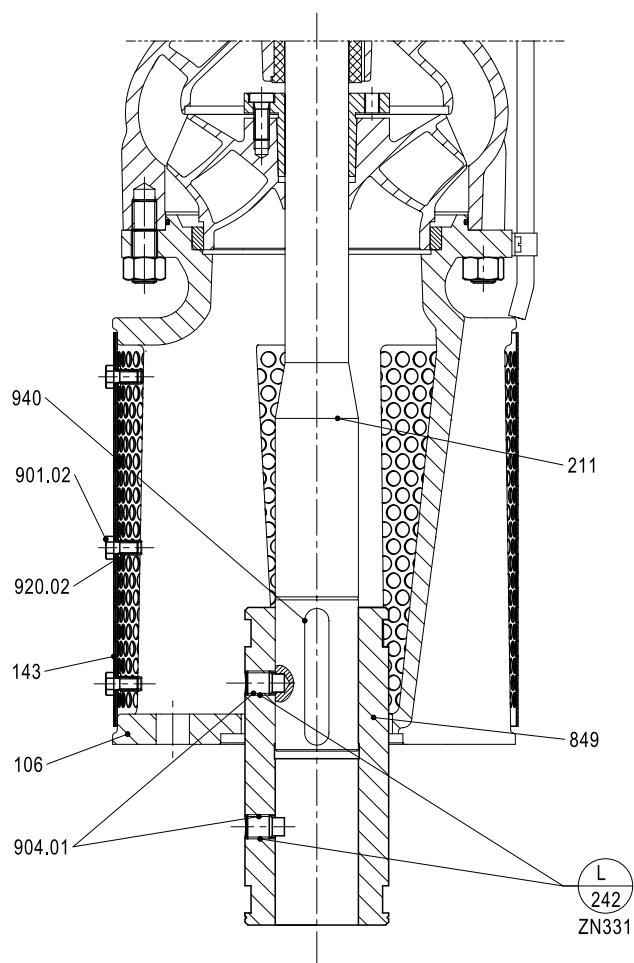


Fig. 10: UPA S 300-270/370 with connection for UMA 300 motor

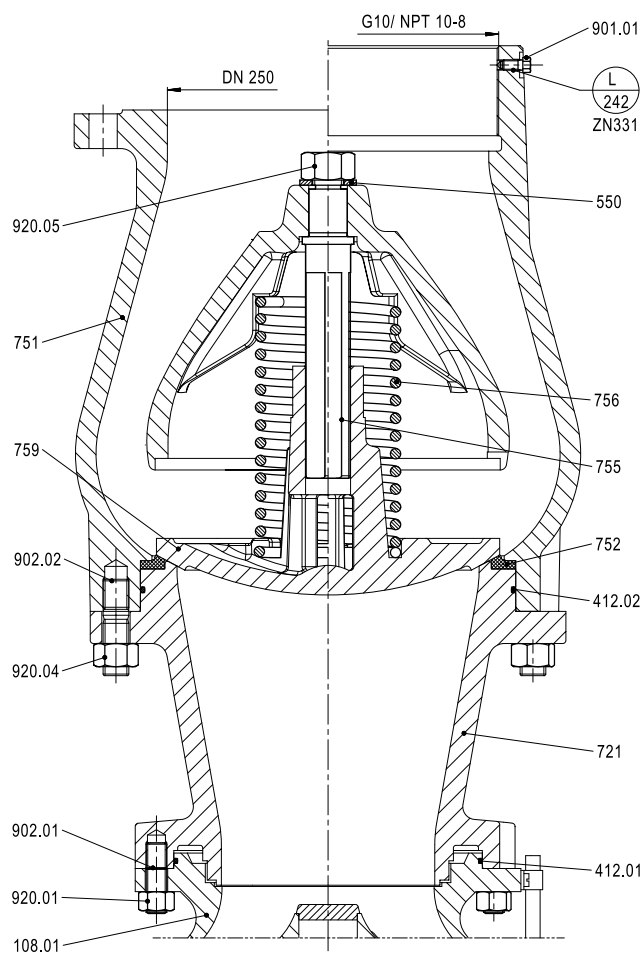


Fig. 11: UPA S 300-370, lift check valve G10/NPT 10-8/DN 250

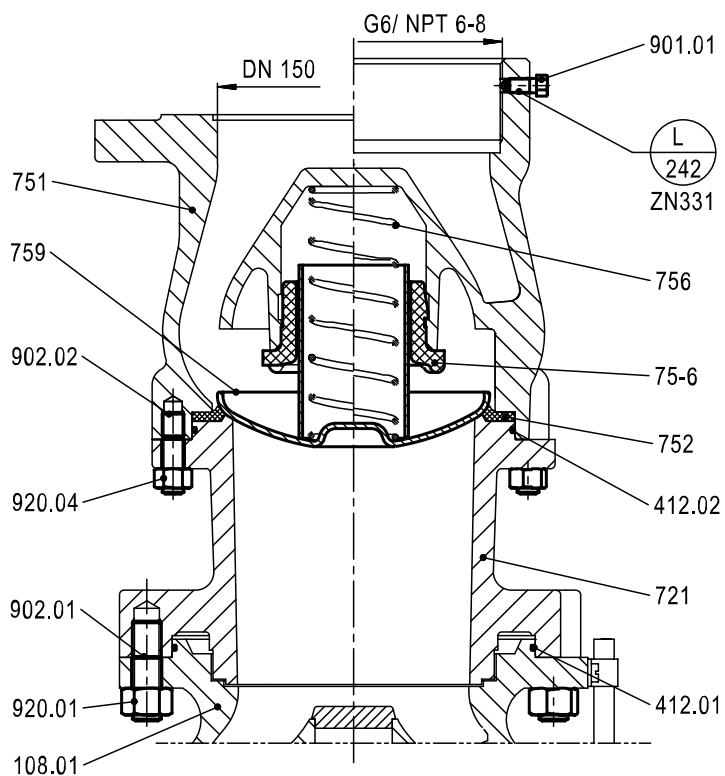


Fig. 12: UPA S 300-270, lift check valve G6/NPT 6-8/DN 150

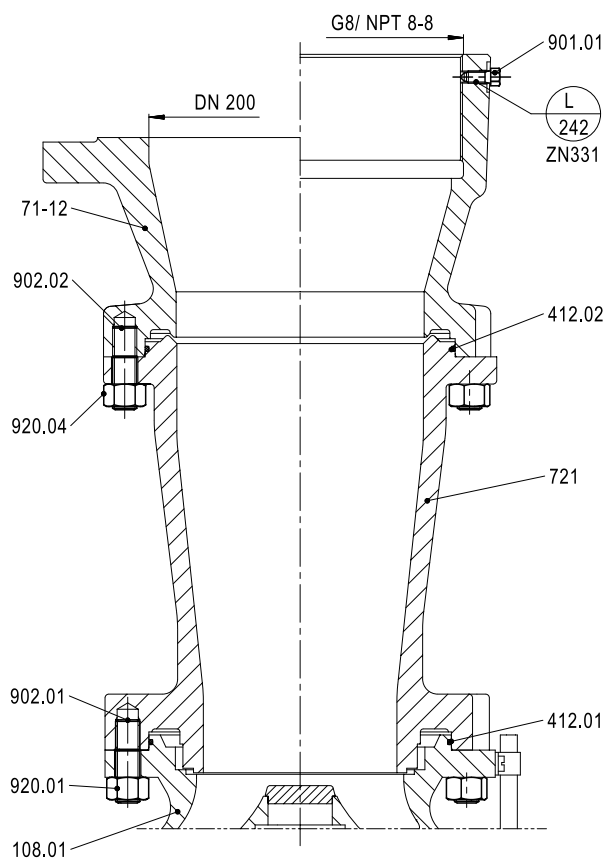


Fig. 13: UPA S 300-270/370, connection nozzle G8/NPT 8-8/DN 200

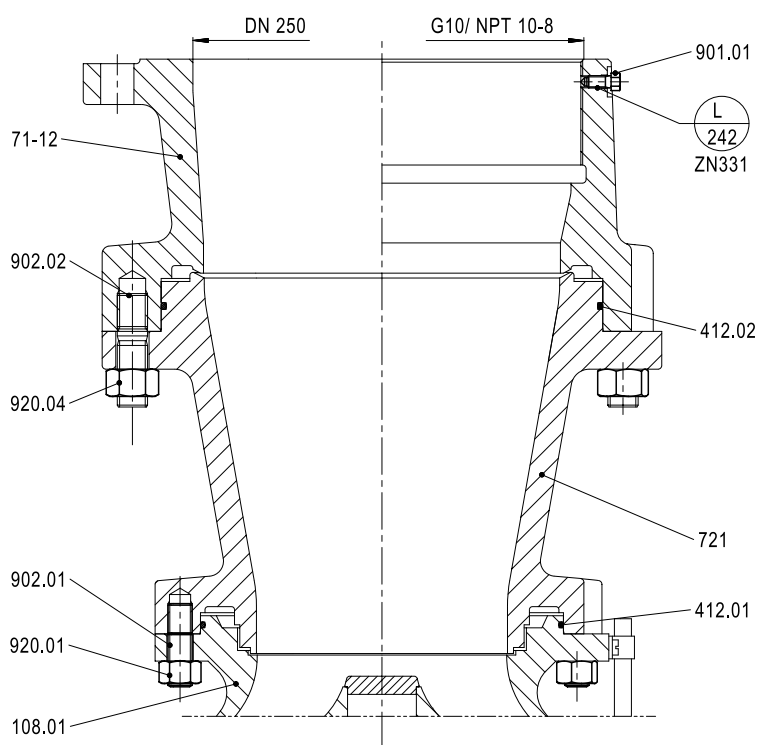


Fig. 14: UPA S 300-370, connection nozzle G10/NPT 10-8/DN 250

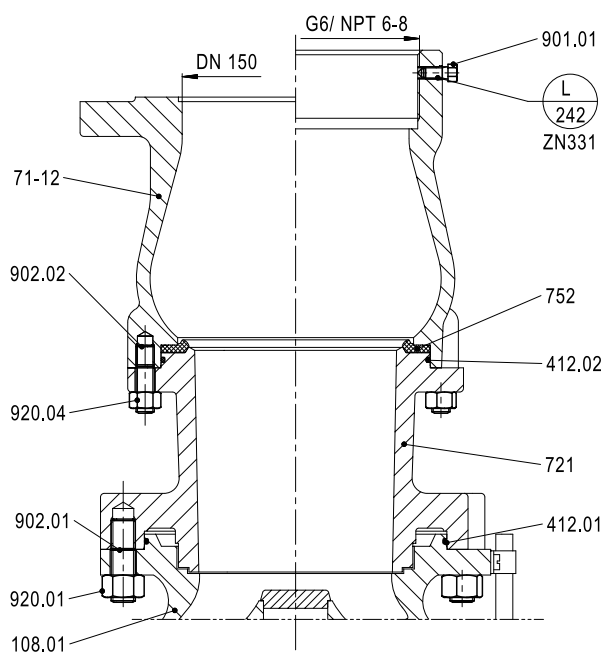


Fig. 15: UPA S 300-270, connection nozzle G6/NPT 6-8/DN 150

Table 28: Symbols key

Symbol	Description
	Always secure screw/bolt/stud connections marked with this symbol with Loctite 242 .
	Always secure connections marked with this symbol with Loctite 648 .

Table 29: List of components for UPA S 300, standard design

Quantity	Part No.	Description	Motor size			Scope of supply of spare parts kit
			8 inch UMA 200, UMA-S 200	10 inch UMA 250, UMA-S 250	12 inch UMA 300	
1	106	Suction casing	✓	-	-	106 + 143, 412.01, 502, 900.02, 901.03, 920.03
			-	✓	-	106 + 143, 412.01, 502, 900.02, 901.03, 920.02, 920.03
			-	-	✓	106 + 143, 412.01, 502, 901.02, 920.02
1	108.01	Stage casing (last stage)	✓	✓	✓	108.01 + 412.01, 540, 58-1, 902.01, 920.01
1 per stage -1	108.02	Stage casing	✓	✓	✓	108.02 + 412.01, 502, 540, 902.01, 920.01
1	143 ²¹⁾	Suction strainer	✓	-	-	143 + 900.02
			-	✓	✓	143 + 901.02, 920.02
1	211	Pump shaft	✓	-	-	211 + 412.03, 540, 849, 904.01, 904.02, 940
			-	✓	✓	211 + 540, 849, 904.01, 940
1 per stage	232	Clockwise impeller	✓	✓	✓	232 + 52-4, 914
1 per stage +1	412.01 ²¹⁾	O-ring	✓	✓	✓	-
1	412.02 ²¹⁾	O-ring	✓	✓	✓	-
1	412.03 ²¹⁾	O-ring	✓	-	-	-
1 per stage	502 ²¹⁾	Casing wear ring	✓	✓	✓	-
1 per stage	52-4 ²¹⁾	Locking sleeve	✓	✓	✓	52-4 + 914
1 per stage	540 ²¹⁾	Bush	✓	✓	✓	-
1	550	Disc	✓	✓	✓	-
1	58-1	Protecting plug	✓	✓	✓	-
1	71-12	Connection nozzle, flanged	✓	✓	✓	71-12 + 752 ²²⁾ , 902.02, 920.04
		Connection nozzle, threaded	✓	✓	✓	71-12 + 752 ²²⁾ , 902.01, 902.02, 920.04
1	721	Adapter	✓	✓	✓	721 + 412.02, 902.01, 920.01
1	75-6 ²¹⁾	Valve disc guide	✓	✓	✓	-
1	751 ²²⁾	Valve body, flanged	✓	✓	✓	751 + 752, 75-6, 756, 759, 902.02, 920.04
		Valve body, threaded	✓	✓	✓	751 + 752, 75-6, 756, 759, 901.01, 902.02, 920.04
	751 ²³⁾	Valve body, flanged	✓	✓	✓	751 + 752, 756, 759, 902.02, 920.04
		Valve body, threaded	✓	✓	✓	751 + 752, 756, 759, 901.01, 902.02, 920.04
	751 ²⁴⁾	Valve body, flanged	✓	✓	✓	751 + 550, 752, 755, 756, 759, 902.02, 920.04, 920.05
		Valve body, threaded	✓	✓	✓	751 + 550, 752, 755, 756, 759, 901.01, 902.02, 920.04, 920.05
1	752 ²¹⁾	Valve seat	✓	✓	✓	-
1	759	Valve disc	✓	✓	✓	759 + 550 ²⁴⁾ , 752, 755 ²⁴⁾ , 756, 920.05 ²⁴⁾
2 per electric cable	81-39	Clamp	✓	✓	✓	-
1 per electric cable	825	Cable guard	✓	✓	✓	-
1	849	Sleeve coupling	✓	-	-	849 + 412.03, 904.01, 904.02, 940
			-	✓	✓	849 + 904.01, 940
4 per electric cable	900.01	Hexagon socket head cap screw	✓	✓	✓	-
3	900.02	Bolt/screw	✓	-	-	-
2	901.01	Hexagon head bolt	✓	✓	✓	-
3	901.02	Hexagon head bolt	✓	✓	-	-
(1 per stage +1) × 12	902.01	Stud	✓	✓	✓	-

21 Recommended spare part

22 Only for G6, NPT6-8 and DN 150 valve bodies

23 Only for G8, NPT8-8 and DN 200 valve bodies

24 Only for G10, NPT10-8 and DN 250 valve bodies

Quantity	Part No.	Description	Motor size			Scope of supply of spare parts kit
			8 inch UMA 200, UMA-S 200	10 inch UMA 250, UMA-S 250	12 inch UMA 300	
12	902.02	Stud	✓	✓	✓	-
1	904.01	Grub screw	✓	-	-	-
4		Grub screw	-	✓	-	-
2		Grub screw	-	-	✓	-
2	904.02	Grub screw	✓	-	-	-
1 per stage × 4	914	Hexagon socket head cap screw	✓	✓	✓	-
(1 per stage +1) × 12	920.01	Nut	✓	✓	✓	-
3	920.02	Nut	-	✓	✓	-
4	920.03	Nut	✓	✓	-	-
12	920.04	Nut	✓	✓	✓	-
1	920.05	Nut	✓	✓	✓	-
1	940 ²¹⁾	Key	✓	✓	✓	-

UPA S 300-270/370, wear-resistant version

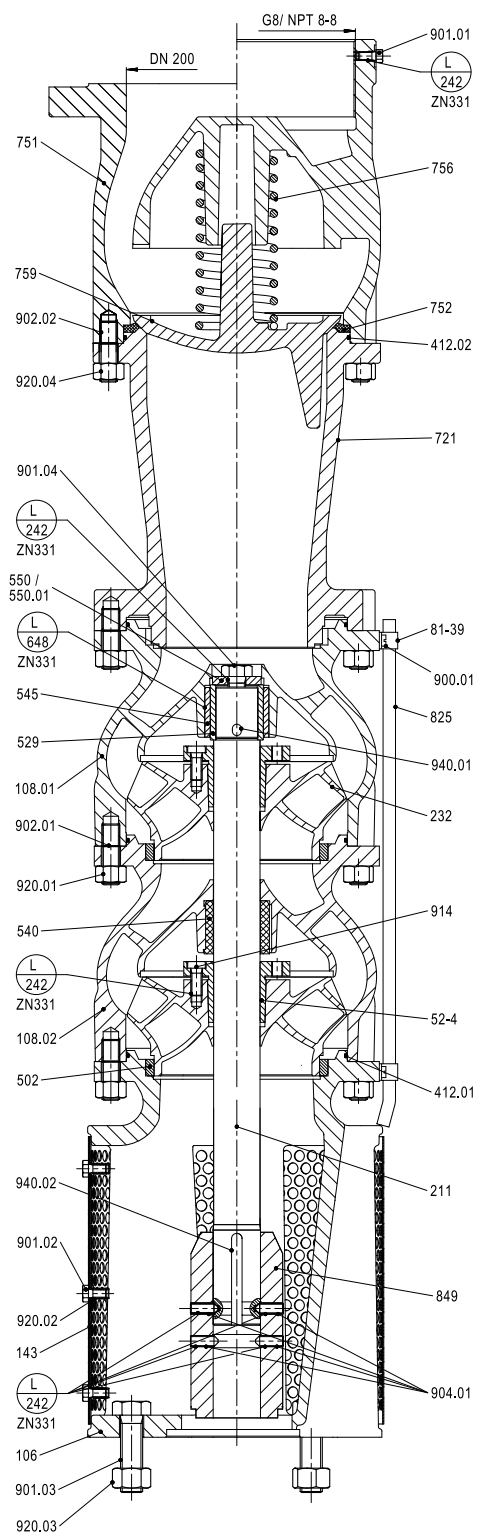


Fig. 16: UPA S 300-270/370 incl. lift check valve G8/NPT 8-8/DN 200, for connection to UMA 250 or UMA-S 250 motor, wear-resistant version

i The following illustrations are identical with those of the standard model:

- UPA S 300-270/370 with connection for UMA 300 motor
- UPA S 300-370 with lift check valve and connection nozzle G10/NPT 10-8/DN 250
- UPA S 300-270 with lift check valve and connection nozzle G6/NPT 6-8/DN 150

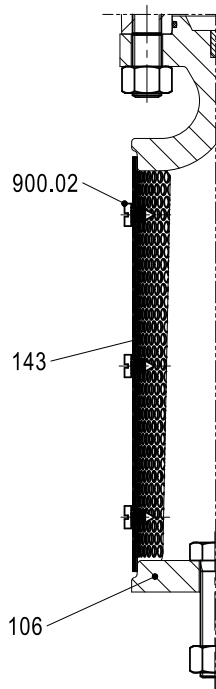


Fig. 17: UPA S 300-270/370, material variant CC, with UMA 200 or UMA S 200, suction strainer, wear-resistant version

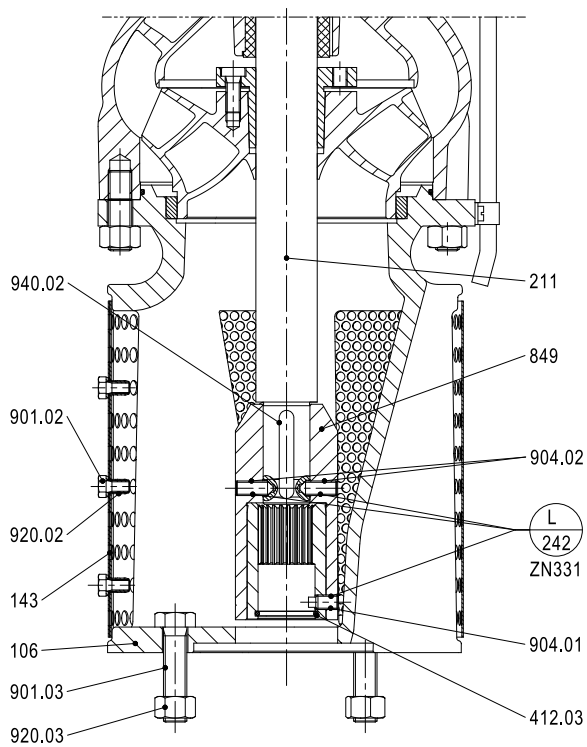


Fig. 18: UPA S 300-270/370 with UMA/UMA S 200, wear-resistant version

Table 30: Symbols key

Symbol	Description
	Always secure screw/bolt/stud connections marked with this symbol with Loctite 242 .
	Always secure connections marked with this symbol with Loctite 648 .

Table 31: List of components for UPA S 300, wear-resistant version

Quantity	Part No.	Description	Motor size			Scope of supply of spare parts kit
			8 inch UMA 200, UMA-S 200	10 inch UMA 250, UMA-S 250	12 inch UMA 300	
1	106	Suction casing	✓	-	-	106 + 143, 412.01, 502, 900.02 ²⁵⁾ , 901.02 ²⁶⁾ 901.03, 920.02 ²⁶⁾ , 920.03
			-	✓	-	106 + 143, 412.01, 502, 901.02, 901.03, 920.02, 920.03
			-	-	✓	106 + 143, 412.01, 502, 901.02, 920.02
1	108.01	Stage casing (last stage)	✓	✓	✓	108.01 + 412.01, 529, 545, 902.01, 920.01
1 per stage -1	108.02	Stage casing	✓	✓	✓	108.02 + 412.01, 502, 540, 902.01, 920.01
1	143 ²⁷⁾	Suction strainer	✓	✓	✓	143 + 900.02 ²⁵⁾ , 901.02 ²⁶⁾ , 920.02 ²⁶⁾
1	211	Pump shaft	✓	-	-	211 + 412.03, 529, 540, 545, 550 ²⁸⁾ / 550.01 ²⁹⁾ , 849, 901.04, 904.01, 904.02, 940.01, 940.02
			-	✓	✓	211 + 529, 540, 545, 550 ²⁸⁾ / 550.01 ²⁹⁾ , 849, 901.04, 904.01, 940.01, 940.02
1 per stage	232	Clockwise impeller	✓	✓	✓	232 + 52-4, 914
1 per stage +1	412.01 ²⁷⁾	O-ring	✓	✓	✓	-
1	412.02 ²⁷⁾	O-ring	✓	✓	✓	-
1	412.03 ²⁷⁾	O-ring	✓	-	-	-
1 per stage	502 ²⁷⁾	Casing wear ring	✓	✓	✓	-
1 per stage	52-4 ²⁷⁾	Locking sleeve	✓	✓	✓	52-4 + 914
1 per stage	540 ²⁷⁾	Bush	✓	✓	✓	-
1	545 ²⁷⁾	Bearing bush	✓	✓	✓	545 + 529
1	550 ²⁸⁾ / 550.01 ²⁹⁾	Disc	✓	✓	✓	-
1	550.02	Disc	✓	✓	✓	-
1	71-12	Connection nozzle, flanged	✓	✓	✓	71-12 + 752 ³⁰⁾ , 902.02, 920.04
		Connection nozzle, threaded	✓	✓	✓	71-12 + 752 ³⁰⁾ , 902.01, 902.02, 920.04
1	721	Adapter	✓	✓	✓	721 + 412.02, 902.01, 920.01
1	75-6 ²⁷⁾	Valve disc guide	✓	✓	✓	-
1	751 ³⁰⁾	Valve body, flanged	✓	✓	✓	751 + 752, 75-6, 756, 759, 902.02, 920.04
		Valve body, threaded	✓	✓	✓	751 + 752, 75-6, 756, 759, 901.01, 902.02, 920.04
	751 ³¹⁾	Valve body, flanged	✓	✓	✓	751 + 752, 756, 759, 902.02, 920.04
		Valve body, threaded	✓	✓	✓	751 + 752, 756, 759, 901.01, 902.02, 920.04
	751 ³²⁾	Valve body, flanged	✓	✓	✓	751 + 550.02, 752, 755, 756, 759, 902.02, 920.04, 920.05
		Valve body, threaded	✓	✓	✓	751 + 550.02, 752, 755, 756, 759, 901.01, 902.02, 920.04, 920.05
1	752 ²⁷⁾	Valve seat (EPDM)	✓	✓	✓	-
1	755	Valve stem	✓	✓	✓	-
1	756 ²⁷⁾	Valve spring	✓	✓	✓	-
1	759	Valve disc	✓	✓	✓	759 + 550.02 ³²⁾ , 752, 755 ³²⁾ , 756, 920.05 ³²⁾

25 Only for material variant C with UMA-S/UMA 200

26 Material variant D only

27 Recommended spare part

28 UPA S 300-270 only

29 UPA S 300-370 only

30 Only for G6, NPT6-8 and DN 150 valve bodies

31 Only for G8, NPT8-8 and DN 200 valve bodies

32 Only for G10, NPT10-8 and DN 250 valve bodies

Quantity	Part No.	Description	Motor size			Scope of supply of spare parts kit
			8 inch UMA 200, UMA-S 200	10 inch UMA 250, UMA-S 250	12 inch UMA 300	
2 per electric cable	81-39	Clamp	✓	✓	✓	-
1 per electric cable	825	Cable guard	✓	✓	✓	-
1	849	Sleeve coupling	✓	-	-	849 + 412.03, 904.01, 904.02, 940.02
			-	✓	✓	849 + 904.01, 940.02
4 per electric cable	900.01	Hexagon socket head cap screw	✓	✓	✓	-
3	900.02 ²⁵⁾	Bolt/screw	✓	-	-	-
2	901.01	Hexagon head bolt	✓	✓	✓	-
3	901.02 ²⁶⁾	Hexagon head bolt	✓	✓	✓	-
4	901.03	Hexagon head bolt	✓	✓	-	-
1	901.04	Hexagon head bolt	✓	✓	✓	-
(1 per stage +1) × 12	902.01	Stud	✓	✓	✓	-
12	902.02	Stud	✓	✓	✓	-
1	904.01	Grub screw	✓	-	-	-
4		Grub screw	-	✓	-	-
2		Grub screw	-	-	✓	-
2	904.02	Grub screw	✓	-	-	-
1 per stage × 4	914	Hexagon socket head cap screw	✓	✓	✓	-
(1 per stage +1) × 12	920.01	Hexagon nut	✓	✓	✓	-
3	920.02 ²⁶⁾	Hexagon nut	-	✓	✓	-
4	920.03	Hexagon nut	✓	✓	-	-
12	920.04	Hexagon nut	✓	✓	✓	-
1	920.05	Hexagon nut	✓	✓	✓	-
1	940.01 ²⁷⁾	Key	✓	✓	✓	-
1	940.02 ²⁷⁾	Key	✓	✓	✓	-

Accessories

Installation parts

Table 32: Selection table

Description			Mat. No.	Weight [kg]
Set of pedestals for horizontal installation, comprising: 2 pedestals and fastening elements				
1.4401	UPA S 300-270 with UMA 200 or UMA-S 200 motor		05100030	7.8
	UPA S 300-270 with UMA 250 or UMA-S 250 motor		05100033	16
	UPA S 300-270 with UMA 300-250/2 motor with support frame		05100044	78.9
	UPA S 300-270 with UMA 300-300/2 motor with support frame		05100045	84.9
	UPA S 300-370 with UMA 200 or UMA-S 200 motor		05100030	7.8
	UPA S 300-370 with UMA 250 or UMA-S 250 motor		05100033	16
	UPA S 300-370 with UMA 300-250/2 motor with support frame		05100044	78.9
	UPA S 300-370 with UMA 300-300/2 motor with support frame		05100045	84.9
1.4539	UPA S 300-270 with UMA 200 or UMA-S 200 motor		05073101	7.8
	UPA S 300-270 with UMA 250 or UMA-S 250 motor		05073104	16
	UPA S 300-270 with UMA 300-250/2 motor with support frame		05073133	78.9
	UPA S 300-270 with UMA 300-300/2 motor with support frame		05073134	84.9
	UPA S 300-370 with UMA 200 or UMA-S 200 motor		05073101	7.8
	UPA S 300-370 with UMA 250 or UMA-S 250 motor		05073104	16
	UPA S 300-370 with UMA 300-250/2 motor with support frame		05073133	78.9
	UPA S 300-370 with UMA 300-300/2 motor with support frame		05073134	84.9
Bolts for fastening the pedestals to the foundation are not included in the scope of supply; they must be provided at the site.				
Suction shroud and pressure shroud			On request	-
Pair of supporting clamps and installation clamps for the following riser sizes, incl. fastening elements, material: painted steel				
R 6 / DN 150	L = 800 mm, F = 31,0 kN		95000304	29
R 8 / DN 200	L = 900 mm, F = 108 kN		95000307	70
R 10 / DN 250	L = 1000 mm, F = 108 kN		95000309	78
2 pairs are required for installation and dismantling.				

Centring device

Table 33: Centring device overview

	Motor size	Well diameter	Mat. No.			Weight
			Material variant			
		[mm]	Plastic	Galvanised steel	Stainless steel	[kg]
	UMA 150	250-299	90047662	-	-	-
	UMA-S 150	300-349	90047663	-	-	-
		350-400	90047664	-	-	-
		UMA 200	300-349	90047665	-	-
	UMA-S 200	350-400	90047666	-	-	-
	UMA-S 250	450-499	-	90037033	90037039	-
	UMA-S 250	500-549	-	90034400	90037041	-
		550-599	-	90037035	90037043	-
		600-650	-	90037037	90037045	-

Cooling shroud
Table 34: Cooling shroud overview

UPA S 300-...	Installation		Motor		Cooling shroud					Strainer				
	Vertical	Horizontal			Ø × length	Weight	Material			Ø × length	Weight	Material		
			1.4301/ AISI304SS	1.4401/ AISI316SS			1.4539/ AISI904L	1.4301/ AISI304SS	1.4401/ AISI316SS			1.4539/ AISI904L		
			Type	P _N [kW]	[mm]	[kg]				[mm]	[kg]			
270	✓	-	UMA 200	37	Ø 380 (400) × 1500	47	01138843	01918320	01918476	Ø 380 × 385	2.4	01138991	01918567	01918579
			UMA 200	55 - 75	Ø 380 (400) × 1800	51	01138846	01918321	01918477	Ø 380 × 385	2.4	01138991	01918567	01918579
			UMA 250	85 - 110	Ø 420 (440) × 1800	56	01138854	01918329	01918485	Ø 420 × 385	4	01138992	01918568	01918580
			UMA 250	132	Ø 420 (440) × 2000	60	01138855	01918330	01918486	Ø 420 × 385	4	01138992	01918568	01918580
			UMA 250	160-190	Ø 420 (440) × 2250	64	01138856	01918331	01918487	Ø 420 × 385	4	01138992	01918568	01918580
			UMA 300	250 - 400	Ø 450 (475) × 2500	71	05139632	-	05139988	Ø 450 × 385	5	01138993	-	01918581
			UMA-S 200	75	Ø 380 (400) × 1500	47	01138843	01918320	01918476	Ø 380 × 385	2.4	01138991	01918567	01918579
			UMA-S 200	100-130	Ø 380 (400) × 1800	51	01138846	01918321	01918477	Ø 380 × 385	2.4	01138991	01918567	01918579
			UMA-S 250	200	Ø 420 (440) × 2000	60	01138855	01918330	01918486	Ø 420 × 385	4	01138992	01918568	01918580
			UMA-S 250	250	Ø 420 (440) × 2250	64	01138856	01918331	01918487	Ø 420 × 385	4	01138992	01918568	01918580
	-	✓	UMA 200	37	Ø 380 (400) × 1500	56.9	01138953	01918376	01918532	Ø 380 × 385	2.4	01138991	01918567	01918579
			UMA 200	55 - 75	Ø 380 (400) × 1800	60.9	01138954	01918377	01918533	Ø 380 × 385	2.4	01138991	01918567	01918579
			UMA 250	85 - 110	Ø 420 (440) × 1800	70.5	01138962	01918385	01918541	Ø 420 × 385	4	01138992	01918568	01918580
			UMA 250	132	Ø 420 (440) × 2000	74.5	01138963	01918386	01918542	Ø 420 × 385	4	01138992	01918568	01918580
			UMA-S 200	75	Ø 380 (400) × 1500	56.9	01138953	01918376	01918532	Ø 380 × 385	2.4	01138991	01918567	01918579
			UMA-S 200	100	Ø 380 (400) × 1800	51	01138846	01918321	01918477	Ø 380 × 385	2.4	01138991	01918567	01918579
			UMA-S 250	200	Ø 420 (440) × 2000	74.5	01138963	01918386	01918542	Ø 420 × 385	4	01138992	01918568	01918580
			370	✓	-	UMA 200	45	Ø 420 (440) × 1500	54	01138851	01918326	01918482	Ø 420 × 385	4
UMA 200	65	Ø 420 (440) × 1800	60			01138852	01918327	01918483	Ø 420 × 385	4	01138992	01918568	01918580	
UMA 200	90	Ø 420 (440) × 2000	62			01138853	01918328	01918484	Ø 420 × 385	4	01138992	01918568	01918580	
UMA 250	85	Ø 420 (440) × 1800	56			01138854	01918329	01918485	Ø 420 × 385	4	01138992	01918568	01918580	
UMA 250	132	Ø 420 (440) × 2000	60			01138855	01918330	01918486	Ø 420 × 385	4	01138992	01918568	01918580	
UMA 250	190	Ø 420 (440) × 2250	64			01138856	01918331	01918487	Ø 420 × 385	4	01138992	01918568	01918580	
UMA 300	250 - 400	Ø 450 (475) × 2500	71			05139632	-	05139988	Ø 450 × 385	5	01138993	-	01918581	
UMA-S 200	75	Ø 420 (440) × 1500	54			01138851	01918326	01918482	Ø 420 × 385	4	01138992	01918568	01918580	
UMA-S 200	100	Ø 420 (440) × 1800	60			01138852	01918327	01918483	Ø 420 × 385	4	01138992	01918568	01918580	
UMA-S 250	200	Ø 420 (440) × 2000	60			01138855	01918330	01918486	Ø 420 × 385	4	01138992	01918568	01918580	
UMA-S 250	250	Ø 420 (440) × 2250	64			01138856	01918331	01918487	Ø 420 × 385	4	01138992	01918568	01918580	
-	✓	UMA 200	45		Ø 420 (440) × 1500	68.5	01138959	01918382	01918538	Ø 420 × 385	4	01138992	01918568	01918580
		UMA 200	65		Ø 420 (440) × 1800	74.5	01138960	01918383	01918539	Ø 420 × 385	4	01138992	01918568	01918580
		UMA 250	85		Ø 420 (440) × 1800	70.5	01138962	01918385	01918541	Ø 420 × 385	4	01138992	01918568	01918580
		UMA 250	132		Ø 420 (440) × 2000	74.5	01138963	01918386	01918542	Ø 420 × 385	4	01138992	01918568	01918580
		UMA-S 200	75		Ø 420 (440) × 1500	68.5	01138959	01918382	01918538	Ø 420 × 385	4	01138992	01918568	01918580

UPA S 300-...	Installation		Motor		Cooling shroud					Strainer				
					Ø × length	Weight	Material			Ø × length	Weight	Material		
	Type	P _N	1.4301/ AISI304SS	1.4401/ AISI316SS			1.4539/ AISI904L	1.4301/ AISI304SS	1.4401/ AISI316SS			1.4539/ AISI904L		
		[kW]			[mm]	[kg]				[mm]	[kg]			
370	-	✓	UMA-S 200	100	Ø 420 (440) × 1800	74.5	01138960	01918383	01918539	Ø 420 × 385	4	01138992	01918568	01918580
			UMA-S 250	200	Ø 420 (440) × 2000	74.5	01138963	01918386	01918542	Ø 420 × 385	4	01138992	01918568	01918580

Flow velocity past the motor

Application

- Required motor cooling

Flow velocity past the motor

$v = 0 \text{ m/s}$

- The flow velocity past the motor is undefined.
- A free flow of heat is established. This heat flow must not be influenced or hindered by any structures, and a supply of fresh water must be provided at all times.
 - Example: pump set freely suspended in a vertical position inside a large tank

Flow velocity past the motor

$v > 0.2 \text{ m/s}$

$v > 0.5 \text{ m/s}$

- The flow velocity past the motor is defined for the installation conditions, see table below. Decisive factors are the inside well diameter or the inside shroud diameter, the operating data and the outside dimensions of the pump set.
 - Example: pump set installed in a vertical position above the screen/filter in a well; values in accordance with the table below.
 - Example: pump set installed in a horizontal position in a tank, fitted with a cooling shroud, or pump set installed in a vertical position in a pump sump, fitted with a cooling shroud; values in accordance with the table below.

Table 35: Maximum permissible well diameter or inside shroud diameter

Flow rate	Flow velocity	Well diameter or inside shroud diameter [mm]				
Q [m³/h]	v [m/s]	UMA 150 UMA-S 150	UMA 200 UMA-S 200	UMA 250 UMA-S 250	UMA 300	14D
15	≥ 0,2	≤ 215	-	-	-	-
	≥ 0,5	≤ 175	-	-	-	-
25	≥ 0,2	≤ 255	-	-	-	-
	≥ 0,5	≤ 195	-	-	-	-
50	≥ 0,2	≤ 330	≤ 350	-	-	-
	≥ 0,5	≤ 235	≤ 265	-	-	-
75	≥ 0,2	≤ 390	≤ 410	≤ 430	-	-
	≥ 0,5	≤ 270	≤ 300	≤ 330	-	-
100	≥ 0,2	≤ 445	≤ 460	≤ 480	-	-
	≥ 0,5	≤ 300	≤ 325	≤ 355	-	-
125	≥ 0,2	≤ 490	≤ 510	≤ 525	-	-
	≥ 0,5	≤ 330	≤ 350	≤ 380	-	-
150	≥ 0,2	≤ 535	≤ 550	≤ 565	≤ 590	-
	≥ 0,5	≤ 355	≤ 380	≤ 400	≤ 430	-
175	≥ 0,2	≤ 575	≤ 590	≤ 605	≤ 625	-
	≥ 0,5	≤ 380	≤ 400	≤ 420	≤ 450	-
200	≥ 0,2	≤ 615	≤ 625	≤ 640	≤ 660	≤ 690
	≥ 0,5	≤ 405	≤ 420	≤ 445	≤ 470	≤ 510
250	≥ 0,2	≤ 680	≤ 690	≤ 705	≤ 725	≤ 750
	≥ 0,5	≤ 445	≤ 460	≤ 480	≤ 505	≤ 540
300	≥ 0,2	≤ 745	≤ 755	≤ 765	≤ 780	≤ 800
	≥ 0,5	≤ 485	≤ 500	≤ 515	≤ 540	≤ 570
350	≥ 0,2	-	≤ 810	≤ 820	≤ 835	≤ 860
	≥ 0,5	-	≤ 530	≤ 550	≤ 570	≤ 600
400	≥ 0,2	-	≤ 865	≤ 875	≤ 890	≤ 910

Flow rate	Flow velocity	Well diameter or inside shroud diameter [mm]				
Q [m³/h]	v [m/s]	UMA 150 UMA-S 150	UMA 200 UMA-S 200	UMA 250 UMA-S 250	UMA 300	14D
400	≥ 0,5	-	≤ 565	≤ 580	≤ 605	≤ 630
500	≥ 0,2	-	≤ 960	≤ 970	≤ 985	≤ 1000
	≥ 0,5	-	≤ 625	≤ 640	≤ 660	≤ 690
600	≥ 0,2	-	≤ 1050	≤ 1055	≤ 1070	≤ 1090
	≥ 0,5	-	≤ 680	≤ 695	≤ 710	≤ 740
800	≥ 0,2	-	≤ 1205	≤ 1215	≤ 1225	≤ 1240
	≥ 0,5	-	≤ 775	≤ 790	≤ 805	≤ 830
1000	≥ 0,2	-	≤ 1345	≤ 1350	≤ 1360	≤ 1370
	≥ 0,5	-	≤ 865	≤ 875	≤ 890	≤ 910
1200	≥ 0,2	-	-	-	≤ 1485	≤ 1500
	≥ 0,5	-	-	-	≤ 965	≤ 980
1400	≥ 0,2	-	-	-	≤ 1600	≤ 1610
	≥ 0,5	-	-	-	≤ 1030	≤ 1050
1600	≥ 0,2	-	-	-	≤ 1705	≤ 1720
	≥ 0,5	-	-	-	≤ 1100	≤ 1120
1800	≥ 0,2	-	-	-	≤ 1805	≤ 1820
	≥ 0,5	-	-	-	≤ 1165	≤ 1180
2000	≥ 0,2	-	-	-	≤ 1900	≤ 1910
	≥ 0,5	-	-	-	≤ 1225	≤ 1240

Power cables

Main applications

- For use with drinking water applications
- Motor lead for submersible motors, laid under water
- Extension cable for the motor lead, laid freely exposed to air and in contact with surfaces
- Unshielded and shielded cable versions

Operating data

Table 36: Operating properties

Characteristic		Value
Rated voltage	V_N [V]	≤ 1000
Ambient temperature	T [°C]	≤ 50
Immersion depth	IM [m]	≤ 500

Designation

Unshielded cable

Example: ZN 1391 - G FL GWT - J 4G25 - Cu-rubber

Table 37: Designation key for unshielded cable

Abbreviation	Description
ZN	Company standard
G	Insulation, rubber
FL	Type of power cable
	FL Flat
	RD Round
GWT	Suitable for drinking water
J	Information about earth conductor
	J With earth conductor
	O Without earth conductor
4G25	Number of cores
	4G25 With earth conductor, 4-core, cross-section 25 mm ²
	3 × 25 Without earth conductor, 3-core, cross-section 25 mm ²
Cu-rubber	Material

Shielded cable

Example: ZN 1391 - S 07 B C4 B - F 3X6 KON

Table 38: Designation key for shielded cable

Abbreviation	Description
ZN	Company standard
S	Basic type according to manufacturer
07	Rated voltage U_0/U
	05 300/500 V
	07 450/750 V
	1/10 600/1000 V
B	Insulation material
	B EPR
C4	Braided shield
B	Sheath material
	B EPR
F	Flexible

Abbreviation	Description
3X6	Nominal cross-section per phase
	3 × 6 Without earth conductor, 3-core, cross-section 6 mm ²
	4G35 With earth conductor, 4-core, cross-section 35 mm ²
KON	Earth conductor design
	KON Earth conductor, concentric as braided shield
	3G16/3 Earth conductor cross-section = nominal phase cross-section (earth conductor divided into three cores, earth conductor cross-section = 16 mm ²), G = with earth conductor

Design details

Unshielded rubber-sheathed cable

Comprising:

- Conductor
 - Copper
 - Bare
 - Fine-stranded
 - Class 5 to DIN EN 60228
- Insulation
 - EPR-based insulation compound (ethylene-propylene rubber) resistant to ozone, weather and water
- Core identification
 - Core colours to DIN EN 50525-1
- Sheath
 - Core insulation made of EPR
 - Outer sheath made of EPDM rubber compound
 - Colour: blue

Shielded rubber-sheathed cable

Comprising:

- Conductor
 - Copper
 - Bare
 - Fine-stranded
 - Class 5 to DIN VDE 0295 / IEC 60228
- Insulation
 - EPR-based insulation compound (ethylene-propylene rubber) resistant to ozone, weather and water
- Core identification
 - Core colours to DIN VDE 0293-308:2003
- Two-layer sheath design
 - Inner sheath: Special EPR compound as per German KTW recommendations, colour: blue
 - Outer sheath: Special EPR compound as per German KTW recommendations, DVGW (Worksheet 270), WRAS and ACS, colour: blue
- Shielding
 - Tinned copper wire braiding over the inner layer of the sheath system, max. coupling resistance 30 Ohm/km at 30 MHz
- Marking
 - HYDROFIRM(T) EMV-FC S1BC4B-F 3X10/10 KON 0,6/1 kV KTW DVGW W270 ACS WRAS

Table 39: Overview of electric cables

Type of electric cable	Characteristic
Flat cable, unshielded	
	3 cores, flat
	4 cores, flat
Round cable, unshielded	
	1 core, round
	4 cores, round
Round cable, shielded	
	4 cores, round

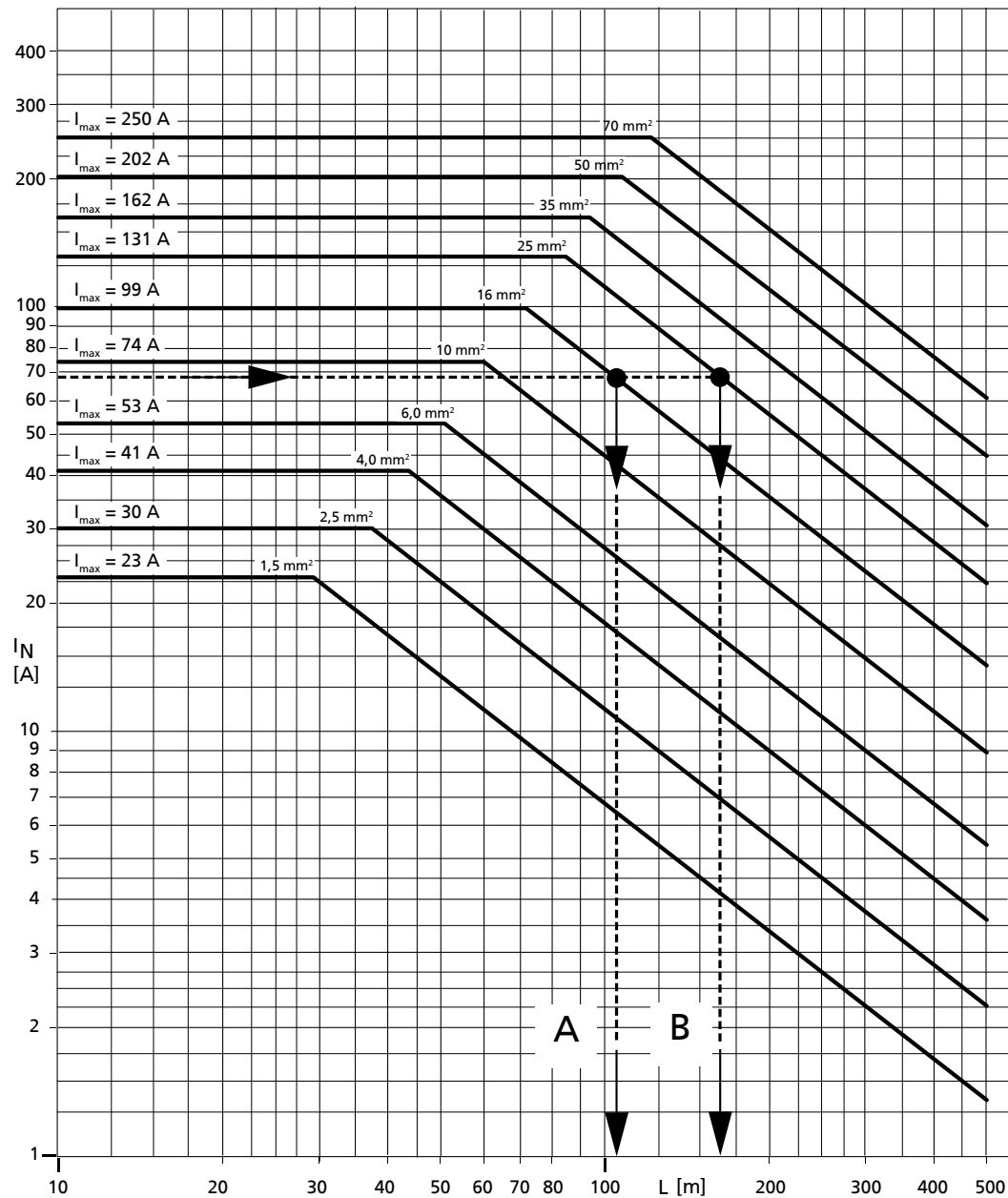
Selection information

Permissible cable lengths for starting method: DOL

Applies to 1 cable or 2 cables in parallel³³⁾

Conditions:

- $V = 400 \text{ V}$; $\Delta V = 3 \%$; $T \leq 30 \text{ °C}$; laid freely exposed to air and in contact with surfaces



Example A

$I_N = 68 \text{ A}$
Cable design: 1 x 16 mm^2
Cable length: $L \leq 105 \text{ m}$

Example B

$I_N = 68 \text{ A}$
Cable design: 1 x 25 mm^2
Cable length: $L \leq 165 \text{ m}$

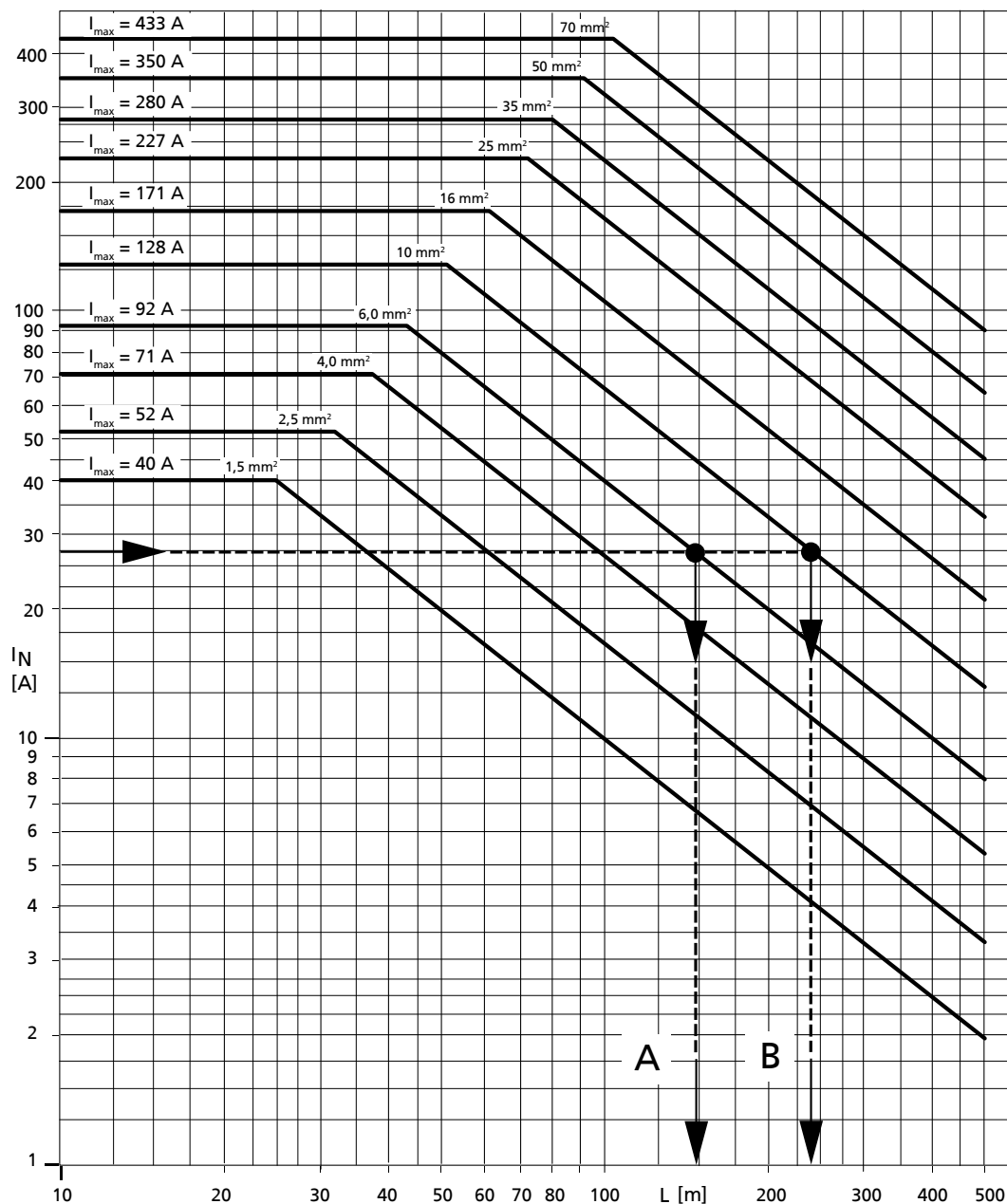
³³ For parallel cables the value is doubled to obtain the permissible length.

Permissible cable lengths for starting method: YΔ

Applies to 2 cables

Conditions:

- $V = 400 \text{ V}$; $\Delta V = 3 \%$; $T \leq 30 \text{ °C}$; laid freely exposed to air and in contact with surfaces



Example A

$I_N = 27 \text{ A}$
Cable design: $2 \times 6.0 \text{ mm}^2$
Cable length: $L \leq 145 \text{ m}$

Example B

$I_N = 27 \text{ A}$
Cable design: $2 \times 10 \text{ mm}^2$
Cable length: $L \leq 235 \text{ m}$

Voltage drop in the extension cable

When calculating the required conductor cross-section q not only I_N , but also the voltage drop ΔV along the cable length L (distance from motor to control unit) must be taken into account. Proper functioning of our submersible motors requires $\Delta V \leq 3 \%$ of the

supply voltage V .

If $\Delta V > 3 \%$, a larger conductor cross-section must be used. The voltage drop is calculated using the following equations:

Starting method DOL/autotransformer

- 1 cable:

$$\Delta U = \frac{3,1 \times L \times I_N \times \cos \varphi}{q \times U} \quad [\%]$$

- 2 cables in parallel (II):

$$\Delta U = \frac{1,55 \times L \times I_N \times \cos \varphi}{q \times U} [\%]$$

Starting method YΔ (2 cables):

$$\Delta U = \frac{2,1 \times L \times I_N \times \cos \varphi}{q \times U} [\%]$$

Power loss ΔP:

$$\Delta P = \frac{\Delta U}{(\cos \varphi)^2} [\%]$$

Table 40: Key

Code	Description
L	Single cable length [m]
I _N	Rated current [A]:
cos φ	Power factor at 4/4 load
q	Conductor cross-section [mm ²]
U	Supply voltage [V]

Maximum permissible rated motor current
Table 41: For ambient temperature t ≤ 30 °C

Starting method	Used as ...	I _{max} [A] for the following conductor cross-sections [mm ²]										
		1,5	2,5	4	6	10	16	25	35	50	70	95
DOL (1 cable or 2 cables in parallel)	Motor lead	29	38	52	67	94	125	166	205	256	316	517
	Extension cable	23	30	41	53	74	99	131	162	202	250	409
YΔ (2 cables)	Motor lead	50	66	90	116	163	217	288	355	443	547	895
	Extension cable	40	52	71	92	128	171	227	280	350	433	708

Dimensions and weights
Table 42: Technical data, unshielded electric extension cable for connection with the motor lead

Core type	Description	Cable cross-section	Height	Width	Diameter	Cable designation	Dimensions	Mat. No.	Weight
		[mm ²]	[mm]	[mm]	[mm]				[kg/m]
●●●	Unshielded electric extension cable, 3-core, flat, without earth conductor	1,5	5,2 ^{+1,0}	11,0 ^{+2,0}	-	G FL GWT-O	3×1,5	90068148	0.11
		2,5	6,1 ^{+1,5}	13,2 ^{+2,3}	-	G FL GWT-O	3×2,5	90068149	0.17
		4	7,0 ^{+2,0}	15,5 ^{+3,5}	-	G FL GWT-O	3×4	90068150	0.25
		6	7,6 ^{+2,2}	17,4 ^{+3,6}	-	G FL GWT-O	3×6	90068151	0.32
		10	9,3 ^{+2,2}	21,5 ^{+3,5}	-	G FL GWT-O	3×10	90068152	0.49
		16	11,2 ^{+2,3}	26,7 ^{+4,3}	-	G FL GWT-O	3×16	90068153	0.75
		25	13,0 ^{+2,5}	31,6 ^{+3,9}	-	G FL GWT-O	3×25	90068154	1.11
		35	14,6 ^{+2,9}	35,5 ^{+5,0}	-	G FL GWT-O	3×35	90068155	1.44
		50	17,0 ^{+3,0}	42,1 ^{+4,9}	-	G FL GWT-O	3×50	90068156	2.05
		70	19,3 ^{+2,7}	48,4 ^{+3,6}	-	G FL GWT-O	3×70	90068157	2.76
		95	-	-	-	-	-	-	-
●●●●	Unshielded electric extension cable, 4-core, flat, with earth conductor	1,5	5,2 ^{+1,0}	14,5 ^{+2,7}	-	G FL GWT-J	4G×1,5	90068158	0.17
		2,5	6,1 ^{+1,5}	17,5 ^{+2,5}	-	G FL GWT-J	4G×2,5	90068159	0.24
		4	-	-	-	G FL GWT-J	4G×4	90068160	0.34
		6	7,6 ^{+2,2}	23,5 ^{+3,0}	-	G FL GWT-J	4G×6	90068161	0.44
		10	9,3 ^{+2,2}	29,0 ^{+3,5}	-	G FL GWT-J	4G×10	90068162	0.7
		16	11,2 ^{+2,3}	35,0 ^{+2,4}	-	G FL GWT-J	4G×16	90068163	1.03
		25	13,0 ^{+2,5}	41,5 ^{+4,5}	-	G FL GWT-J	4G×25	90068164	1.46
		35	-	-	-	G FL GWT-J	4G×25	90068165	1.96
		50	-	-	-	-	-	-	-
		70	-	-	-	-	-	-	-
		95	-	-	-	-	-	-	-
●	Unshielded electric extension cable, 1-core, round, without earth conductor	1,5	-	-	5,3 ^{+1,1}	G RD GWT-O	1×1,5	01046306	0.01
		2,5	-	-	-	-	-	-	-
		4	-	-	-	-	-	-	-
		6	-	-	7,4 ^{+1,2}	G RD GWT-O	1×6	01042783	0.11

Core type	Description	Cable cross-section	Height	Width	Diameter	Cable designation	Dimensions	Mat. No.	Weight
		[mm²]							
	Unshielded electric extension cable, 1-core, round, without earth conductor	10	-	-	9,1 ^{+3,4}	G RD GWT-O	1x10	01042784	0.18
		16	-	-	10,7 ^{+2,4}	G RD GWT-O	1x16	01042785	0.27
		25	-	-	12,6 ^{+2,6}	G RD GWT-O	1x25	01042786	0.38
		35	-	-	13,8 ^{+3,6}	G RD GWT-O	1x35	01042787	0.5
		50	-	-	16,2 ^{+3,6}	G RD GWT-O	1x50	01042788	0.7
		70	-	-	18,5 ^{+3,6}	G RD GWT-O	1x70	01042789	0.94
		95	-	-	21,9 ^{+1,5}	G RD GWT-O	1x95	01042794	1.15
	Unshielded electric extension cable, 4-core, round, with earth conductor	1,5	-	-	10,0 ^{+2,0}	G RD GWT-J	4G1,5	90068174	0.18
		2,5	-	-	12,0 ^{+1,9}	G RD GWT-J	4G2,5	90068175	0.26
		4	-	-	13,9 ^{+2,0}	G RD GWT-J	4G4	90068176	0.36
		6	-	-	15,7 ^{+2,1}	G RD GWT-J	4G6	90068177	0.48
		10	-	-	21,1 ^{+2,1}	G RD GWT-J	4G10	90068178	0.84
		16	-	-	24,5 ^{+4,3}	G RD GWT-J	4G16	90068179	1.22
		25	-	-	29,7 ^{+4,3}	G RD GWT-J	4G25	90068180	1.77
		35	-	-	33,3 ^{+5,5}	G RD GWT-J	4G35	90068181	2.3
		50	-	-	39,0 ^{+5,6}	G RD GWT-J	4G50	90068182	3.19
		70	-	-	44,2 ^{+5,8}	G RD GWT-J	4G70	90068183	4.36
		95	-	-	-	-	-	-	-

Table 43: Technical data, shielded electric extension cable for connection with the motor lead

Core type	Description	Cable cross-section		Cable designation	Dimensions	Mat. No.	Weight
		[mm²]	Diameter [mm]				
	Shielded electric extension cable, 4-core, round	0,5	7,5 ^{+1,0}	S 05 B C4 B-F	F4x0,5	05536014	0.09
		6	17,8 ^{+2,0}	S 07 B C4 B-F	3x6/6 KON	05535908	0.5
		16	24,7 ^{+3,0}	S 07 B C4 B-F	3x16+3G16/3	05535909	1.25
		35	32,1 ^{+2,0}	S 07 B C4 B-F	3x35+3G16/3	05536010	2.1
		50	36,8 ^{+2,0}	S 07 B C4 B-F	3x50+3G25/3	05536011	2.78
		70	40,4 ^{+3,0}	S 07 B C4 B-F	3x70+3G35/3	05536012	3.77
		95	45,0 ^{+2,0}	S 07 B C4 B-F	3x95+3G50/3	05536013	4.85

Cable connector

Application

For providing a sealed, water-tight connection between electric extension cables and motor leads of submersible motors.

- Ambient temperature: $T \leq +50\text{ °C}$
- Immersion depth: $\leq 500\text{ m}$

Selection table

The table below only shows the **electric cables on stock**. Cable connectors for other electric cables on request

Table 44: Key

Cable symbols	Explanation	Cable designation
	1-core, round, unshielded	G RD GWT - O 1 G
	4-core, round, unshielded	G RD GWT - J 4 G
	3-core, flat, unshielded	G FL GWT - O 3x ...
	4-core, flat, unshielded	G FL GWT - J 4G ...
	4-core, round, shielded	S 07 B C4 B-F - 4G35

Cable connector for an electric extension cable
Table 45: Cable connector: electric extension cable, →

Motor lead	Electric extension cable [mm²]										
[mm²]	1,5	2,5	4	6	10	16	25	35	50	70	95
1,5	28 90049385 ⁽³⁴⁾ 95005106 ⁽³⁵⁾	28 90049385 ⁽³⁴⁾ 95005106 ⁽³⁵⁾	28 90049385 ⁽³⁴⁾ 95005106 ⁽³⁵⁾	35 90049387 ⁽³⁴⁾ 90049397 ⁽³⁵⁾	35 90049387 ⁽³⁴⁾ 90049397 ⁽³⁵⁾	43 90049389 ⁽³⁴⁾ 90049399 ⁽³⁵⁾	53 90049391 ⁽³⁴⁾ 90049401 ⁽³⁵⁾	53 90049391 ⁽³⁴⁾ 90049401 ⁽³⁵⁾	-	-	-
2,5	28 90049385 ⁽³⁴⁾ 95005106 ⁽³⁵⁾	28 90049385 ⁽³⁴⁾ 95005106 ⁽³⁵⁾	28 90049385 ⁽³⁴⁾ 95005106 ⁽³⁵⁾	35 90049387 ⁽³⁴⁾ 90049397 ⁽³⁵⁾	35 90049387 ⁽³⁴⁾ 90049397 ⁽³⁵⁾	43 90049389 ⁽³⁴⁾ 90049399 ⁽³⁵⁾	53 90049391 ⁽³⁴⁾ 90049401 ⁽³⁵⁾	53 90049391 ⁽³⁴⁾ 90049401 ⁽³⁵⁾	-	-	-
4	28 90049385 ⁽³⁴⁾ 95005106 ⁽³⁵⁾	28 90049385 ⁽³⁴⁾ 95005106 ⁽³⁵⁾	28 90049385 ⁽³⁴⁾ 95005106 ⁽³⁵⁾	35 90049387 ⁽³⁴⁾ 90049397 ⁽³⁵⁾	35 90049387 ⁽³⁴⁾ 90049397 ⁽³⁵⁾	43 90049389 ⁽³⁴⁾ 90049399 ⁽³⁵⁾	53 90049391 ⁽³⁴⁾ 90049401 ⁽³⁵⁾	53 90049391 ⁽³⁴⁾ 90049401 ⁽³⁵⁾	-	-	-
6	-	35 90049387 ⁽³⁴⁾ 90049397 ⁽³⁵⁾	35 90049387 ⁽³⁴⁾ 90049397 ⁽³⁵⁾	35 90049387 ⁽³⁴⁾ 90049397 ⁽³⁵⁾	35 90049387 ⁽³⁴⁾ 90049397 ⁽³⁵⁾	43 90049389 ⁽³⁴⁾ 90049399 ⁽³⁵⁾	53 90049391 ⁽³⁴⁾ 90049401 ⁽³⁵⁾	53 90049391 ⁽³⁴⁾ 90049401 ⁽³⁵⁾	-	-	-
	-	-	-	-	-	29F 95005160 ⁽³⁴⁾	29F 95005160 ⁽³⁴⁾	39F 95005510 ⁽³⁴⁾	-	-	-
10	-	35 90049387 ⁽³⁴⁾ 90049397 ⁽³⁵⁾	35 90049387 ⁽³⁴⁾ 90049397 ⁽³⁵⁾	43 90049389 ⁽³⁴⁾ 90053910 ⁽³⁵⁾	43 90049389 ⁽³⁴⁾ 90053910 ⁽³⁵⁾	43 90049389 ⁽³⁴⁾ 90053910 ⁽³⁵⁾	53 90049391 ⁽³⁴⁾ 90049401 ⁽³⁵⁾	53 90049391 ⁽³⁴⁾ 90049401 ⁽³⁵⁾	66 95005523 ⁽³⁴⁾ 90053914 ⁽³⁵⁾	-	-
	-	-	-	-	29F 95005159 ⁽³⁴⁾	29F 95005161 ⁽³⁴⁾	29F 95005161 ⁽³⁴⁾	39F 95005176 ⁽³⁴⁾	-	-	-
16	-	43 90049389 ⁽³⁴⁾ 90049399 ⁽³⁵⁾	43 90049389 ⁽³⁴⁾ 90049399 ⁽³⁵⁾	43 90049389 ⁽³⁴⁾ 90049399 ⁽³⁵⁾	43 90049389 ⁽³⁴⁾ 90053910 ⁽³⁵⁾	53 95005192 ⁽³⁴⁾ 90053913 ⁽³⁵⁾	53 95005192 ⁽³⁴⁾ 90053913 ⁽³⁵⁾	53 95005192 ⁽³⁴⁾ 90053913 ⁽³⁵⁾	66 95005523 ⁽³⁴⁾ 90053914 ⁽³⁵⁾	66 95005523 ⁽³⁴⁾ 90053914 ⁽³⁵⁾	-
25	-	53 90049391 ⁽³⁴⁾ 90049401 ⁽³⁵⁾	53 90049391 ⁽³⁴⁾ 90049401 ⁽³⁵⁾	53 90049391 ⁽³⁴⁾ 90049401 ⁽³⁵⁾	53 90049391 ⁽³⁴⁾ 90049401 ⁽³⁵⁾	53 95005192 ⁽³⁴⁾ 90053913 ⁽³⁵⁾	66 95005214 ⁽³⁴⁾ 90053914 ⁽³⁵⁾	66 95005214 ⁽³⁴⁾ 90053914 ⁽³⁵⁾	66 95005523 ⁽³⁴⁾ 90053914 ⁽³⁵⁾	66 95005523 ⁽³⁴⁾ 90053914 ⁽³⁵⁾	-
35	-	53 90049391 ⁽³⁴⁾ 90049401 ⁽³⁵⁾	53 90049391 ⁽³⁴⁾ 90049401 ⁽³⁵⁾	53 90049391 ⁽³⁴⁾ 90049401 ⁽³⁵⁾	53 90049391 ⁽³⁴⁾ 90049401 ⁽³⁵⁾	53 95005192 ⁽³⁴⁾ 90053913 ⁽³⁵⁾	66 95005214 ⁽³⁴⁾ 90053914 ⁽³⁵⁾	66 95005206 ⁽³⁴⁾ 90053907 ⁽³⁵⁾	66 95005206 ⁽³⁴⁾ 90053907 ⁽³⁵⁾	66 95005206 ⁽³⁴⁾ 90053907 ⁽³⁵⁾	-
50	-	-	-	-	-	66 95005523 ⁽³⁴⁾ 90053914 ⁽³⁵⁾	66 95005523 ⁽³⁴⁾ 90053914 ⁽³⁵⁾	66 95005206 ⁽³⁴⁾ 90053907 ⁽³⁵⁾	66 95005206 ⁽³⁴⁾ 90053907 ⁽³⁵⁾	66 95005206 ⁽³⁴⁾ 90053907 ⁽³⁵⁾	-
70	-	-	-	-	-	66 95005523 ⁽³⁴⁾ 90053914 ⁽³⁵⁾	66 95005523 ⁽³⁴⁾ 90053914 ⁽³⁵⁾	66 95005206 ⁽³⁴⁾ 90053907 ⁽³⁵⁾	66 95005206 ⁽³⁴⁾ 90053907 ⁽³⁵⁾	66 95005207 ⁽³⁴⁾ 90053907 ⁽³⁵⁾	-

Table 46: Cable connector: electric extension cable, →

Motor lead	Electric extension cable [mm²]										
[mm²]	1,5	2,5	4	6	10	16	25	35	50	70	95
1,5	28 90049385 ⁽³⁴⁾ 95005106 ⁽³⁵⁾	28 90049385 ⁽³⁴⁾ 95005106 ⁽³⁵⁾	28 90049385 ⁽³⁴⁾ 95005106 ⁽³⁵⁾	35 90049387 ⁽³⁴⁾ 90049397 ⁽³⁵⁾	35 90049387 ⁽³⁴⁾ 90049397 ⁽³⁵⁾	43 90049389 ⁽³⁴⁾ 90049399 ⁽³⁵⁾	53 90049391 ⁽³⁴⁾ 90049401 ⁽³⁵⁾	53 90049391 ⁽³⁴⁾ 90049401 ⁽³⁵⁾	-	-	-
2,5	28 90049385 ⁽³⁴⁾ 95005106 ⁽³⁵⁾	28 90049385 ⁽³⁴⁾ 95005106 ⁽³⁵⁾	28 90049385 ⁽³⁴⁾ 95005106 ⁽³⁵⁾	35 90049387 ⁽³⁴⁾ 90049397 ⁽³⁵⁾	35 90049387 ⁽³⁴⁾ 90049397 ⁽³⁵⁾	43 90049389 ⁽³⁴⁾ 90049399 ⁽³⁵⁾	53 90049391 ⁽³⁴⁾ 90049401 ⁽³⁵⁾	53 90049391 ⁽³⁴⁾ 90049401 ⁽³⁵⁾	-	-	-
4	28 90049385 ⁽³⁴⁾ 95005106 ⁽³⁵⁾	35 90049387 ⁽³⁴⁾ 90049397 ⁽³⁵⁾	35 90049387 ⁽³⁴⁾ 90049397 ⁽³⁵⁾	35 90049387 ⁽³⁴⁾ 90049397 ⁽³⁵⁾	35 90049387 ⁽³⁴⁾ 90049397 ⁽³⁵⁾	43 90049389 ⁽³⁴⁾ 90049399 ⁽³⁵⁾	53 90049391 ⁽³⁴⁾ 90049401 ⁽³⁵⁾	53 90049391 ⁽³⁴⁾ 90049401 ⁽³⁵⁾	-	-	-

34 MVB - material No. for cable connector, extended and sealed during manufacture

35 OVB - material No. for cable connector, not connected and supplied but not fitted

Motor lead	Electric extension cable [mm ²]										
[mm ²]	1,5	2,5	4	6	10	16	25	35	50	70	95
6	35 90049387 ⁽³⁴⁾ 90049397 ⁽³⁵⁾	35 90049387 ⁽³⁴⁾ 90049397 ⁽³⁵⁾	35 90049387 ⁽³⁴⁾ 90049397 ⁽³⁵⁾	35 90049387 ⁽³⁴⁾ 90049397 ⁽³⁵⁾	35 90049387 ⁽³⁴⁾ 90049397 ⁽³⁵⁾	43 90049389 ⁽³⁴⁾ 90049399 ⁽³⁵⁾	53 90049391 ⁽³⁴⁾ 90049401 ⁽³⁵⁾	53 90049391 ⁽³⁴⁾ 90049401 ⁽³⁵⁾	-	-	-
10	43 90049389 ⁽³⁴⁾ 90053910 ⁽³⁵⁾	43 90049389 ⁽³⁴⁾ 90053910 ⁽³⁵⁾	43 90049389 ⁽³⁴⁾ 90053910 ⁽³⁵⁾	43 90049389 ⁽³⁴⁾ 90053910 ⁽³⁵⁾	43 90049389 ⁽³⁴⁾ 90053910 ⁽³⁵⁾	43 90049389 ⁽³⁴⁾ 90053910 ⁽³⁵⁾	53 90049391 ⁽³⁴⁾ 90049401 ⁽³⁵⁾	53 90049391 ⁽³⁴⁾ 90049401 ⁽³⁵⁾	66 95005523 ⁽³⁴⁾ 90053914 ⁽³⁵⁾	-	-
16	43 90049389 ⁽³⁴⁾ 90049399 ⁽³⁵⁾	43 90049389 ⁽³⁴⁾ 90049399 ⁽³⁵⁾	43 90049389 ⁽³⁴⁾ 90049399 ⁽³⁵⁾	43 90049389 ⁽³⁴⁾ 90049399 ⁽³⁵⁾	43 90049389 ⁽³⁴⁾ 90053910 ⁽³⁵⁾	53 95005192 ⁽³⁴⁾ 90053913 ⁽³⁵⁾	53 95005192 ⁽³⁴⁾ 90053913 ⁽³⁵⁾	53 95005192 ⁽³⁴⁾ 90053913 ⁽³⁵⁾	66 95005523 ⁽³⁴⁾ 90053914 ⁽³⁵⁾	66 95005523 ⁽³⁴⁾ 90053914 ⁽³⁵⁾	-
25	53 90049391 ⁽³⁴⁾ 90049401 ⁽³⁵⁾	53 90049391 ⁽³⁴⁾ 90049401 ⁽³⁵⁾	53 90049391 ⁽³⁴⁾ 90049401 ⁽³⁵⁾	53 90049391 ⁽³⁴⁾ 90049401 ⁽³⁵⁾	53 90049391 ⁽³⁴⁾ 90049401 ⁽³⁵⁾	53 95005192 ⁽³⁴⁾ 90053913 ⁽³⁵⁾	66 95005214 ⁽³⁴⁾ 90053914 ⁽³⁵⁾	66 95005214 ⁽³⁴⁾ 90053914 ⁽³⁵⁾	66 95005523 ⁽³⁴⁾ 90053914 ⁽³⁵⁾	66 95005523 ⁽³⁴⁾ 90053914 ⁽³⁵⁾	-

Table 47: Cable connector: electric extension cable, + or + →

Motor lead	Electric extension cable [mm ²]										
[mm ²]	1,5	2,5	4	6	10	16	25	35	50	70	95
+ or +											
1,5	28 90049386 ⁽³⁴⁾ 90049396 ⁽³⁵⁾	28 90049386 ⁽³⁴⁾ 90049396 ⁽³⁵⁾	28 90049386 ⁽³⁴⁾ 90049396 ⁽³⁵⁾	35 95005165 ⁽³⁴⁾ 90039349 ⁽³⁵⁾	35 95005165 ⁽³⁴⁾ 90039349 ⁽³⁵⁾	-	-	-	-	-	-
2,5	28 90049388 ⁽³⁴⁾ 90049398 ⁽³⁵⁾	28 90049388 ⁽³⁴⁾ 90049398 ⁽³⁵⁾	28 90049388 ⁽³⁴⁾ 90049398 ⁽³⁵⁾	35 95005165 ⁽³⁴⁾ 90039349 ⁽³⁵⁾	35 95005165 ⁽³⁴⁾ 90039349 ⁽³⁵⁾	-	-	-	-	-	-
4	-	-	35 95005165 ⁽³⁴⁾ 90039349 ⁽³⁵⁾	35 95005165 ⁽³⁴⁾ 90039349 ⁽³⁵⁾	35 95005165 ⁽³⁴⁾ 90039349 ⁽³⁵⁾	43 90049390 ⁽³⁴⁾ 90049400 ⁽³⁵⁾	-	-	-	-	-
6	-	-	-	43 90049390 ⁽³⁴⁾ 90049400 ⁽³⁵⁾	43 90049390 ⁽³⁴⁾ 90049400 ⁽³⁵⁾	43 90049390 ⁽³⁴⁾ 90049400 ⁽³⁵⁾	53 90049392 ⁽³⁴⁾ 90049402 ⁽³⁵⁾	53 90049392 ⁽³⁴⁾ 90049402 ⁽³⁵⁾	66 90033248 ⁽³⁴⁾ 90055737 ⁽³⁵⁾	66 90033248 ⁽³⁴⁾ 90055737 ⁽³⁵⁾	-
10	-	-	-	66 95005522 ⁽³⁴⁾ 90054294 ⁽³⁵⁾	66 95005522 ⁽³⁴⁾ 90054294 ⁽³⁵⁾	66 90039412 ⁽³⁴⁾ 90053905 ⁽³⁵⁾	66 90039412 ⁽³⁴⁾ 90053905 ⁽³⁵⁾	66 90039412 ⁽³⁴⁾ 90053905 ⁽³⁵⁾	66 90039412 ⁽³⁴⁾ 90053905 ⁽³⁵⁾	66 90033248 ⁽³⁴⁾ 90055737 ⁽³⁵⁾	-
16	-	-	-	-	-	66 90039361 ⁽³⁴⁾ 90053906 ⁽³⁵⁾	66 90039361 ⁽³⁴⁾ 90053906 ⁽³⁵⁾	66 90039361 ⁽³⁴⁾ 90053906 ⁽³⁵⁾	66 90036538 ⁽³⁴⁾ 90055738 ⁽³⁵⁾	66 90036538 ⁽³⁴⁾ 90055738 ⁽³⁵⁾	-
25	-	-	-	-	-	-	78 90039701 ⁽³⁴⁾ 90053908 ⁽³⁵⁾	78 90039701 ⁽³⁴⁾ 90053908 ⁽³⁵⁾	78 90039701 ⁽³⁴⁾ 90053908 ⁽³⁵⁾	78 90039701 ⁽³⁴⁾ 90053908 ⁽³⁵⁾	-

Table 48: Cable connector: electric extension cable, →

Motor lead	Electric extension cable [mm ²]										
[mm ²]	1,5	2,5	4	6	10	16	25	35	50	70	95
1,5	28 90049385 ³⁴⁾ 95005106 ³⁵⁾	28 90049385 ³⁴⁾ 95005106 ³⁵⁾	28 90049385 ³⁴⁾ 95005106 ³⁵⁾	28 90049385 ³⁴⁾ 95005106 ³⁵⁾	35 90049387 ³⁴⁾ 90049397 ³⁵⁾	43 90049389 ³⁴⁾ 90049399 ³⁵⁾	-	-	-	-	-
2,5	28 90049385 ³⁴⁾ 95005106 ³⁵⁾	28 90049385 ³⁴⁾ 95005106 ³⁵⁾	28 90049385 ³⁴⁾ 95005106 ³⁵⁾	28 90049385 ³⁴⁾ 95005106 ³⁵⁾	35 90049387 ³⁴⁾ 90049397 ³⁵⁾	43 90049389 ³⁴⁾ 90049399 ³⁵⁾	-	-	-	-	-
4	28 90049385 ³⁴⁾ 95005106 ³⁵⁾	28 90049385 ³⁴⁾ 95005106 ³⁵⁾	28 90049385 ³⁴⁾ 95005106 ³⁵⁾	28 90049385 ³⁴⁾ 95005106 ³⁵⁾	35 90049387 ³⁴⁾ 90049397 ³⁵⁾	43 90049389 ³⁴⁾ 90049399 ³⁵⁾	-	-	-	-	-
6	28 90049385 ³⁴⁾ 95005106 ³⁵⁾	28 90049385 ³⁴⁾ 95005106 ³⁵⁾	28 90049385 ³⁴⁾ 95005106 ³⁵⁾	28 90049385 ³⁴⁾ 95005106 ³⁵⁾	35 90049387 ³⁴⁾ 90049397 ³⁵⁾	43 90049389 ³⁴⁾ 90049399 ³⁵⁾	53 90049391 ³⁴⁾ 90049401 ³⁵⁾	53 90049391 ³⁴⁾ 90049401 ³⁵⁾	-	-	-
10	35 90049387 ³⁴⁾ 90049397 ³⁵⁾	35 90049387 ³⁴⁾ 90049397 ³⁵⁾	35 90049387 ³⁴⁾ 90049397 ³⁵⁾	35 90049387 ³⁴⁾ 90049397 ³⁵⁾	35 90049387 ³⁴⁾ 90049397 ³⁵⁾	43 90049389 ³⁴⁾ 90049399 ³⁵⁾	53 90049391 ³⁴⁾ 90049401 ³⁵⁾	53 90049391 ³⁴⁾ 90049401 ³⁵⁾	66 95005523 ³⁴⁾ 90053914 ³⁵⁾	-	-
16	43 90049389 ³⁴⁾ 90049399 ³⁵⁾	43 90049389 ³⁴⁾ 90049399 ³⁵⁾	43 90049389 ³⁴⁾ 90049399 ³⁵⁾	43 90049389 ³⁴⁾ 90049399 ³⁵⁾	43 90049389 ³⁴⁾ 90049399 ³⁵⁾	43 90049389 ³⁴⁾ 90049399 ³⁵⁾	53 90049391 ³⁴⁾ 90049401 ³⁵⁾	53 90049391 ³⁴⁾ 90049401 ³⁵⁾	66 95005370 ³⁴⁾ 90053914 ³⁵⁾	66 95005214 ³⁴⁾ 90053914 ³⁵⁾	-
25	-	-	-	53 90049391 ³⁴⁾ 90049401 ³⁵⁾	53 90049391 ³⁴⁾ 90049401 ³⁵⁾	53 90049391 ³⁴⁾ 90049401 ³⁵⁾	53 90049391 ³⁴⁾ 90049401 ³⁵⁾	53 90049391 ³⁴⁾ 90049401 ³⁵⁾	66 95005520 ³⁴⁾ 90053914 ³⁵⁾	78 95005527 ³⁴⁾	-
35	-	-	-	53 90049391 ³⁴⁾ 90049401 ³⁵⁾	53 90049391 ³⁴⁾ 90049401 ³⁵⁾	53 90049391 ³⁴⁾ 90049401 ³⁵⁾	53 90049391 ³⁴⁾ 90049401 ³⁵⁾	53 90049391 ³⁴⁾ 90049401 ³⁵⁾	66 95005214 ³⁴⁾ 90053914 ³⁵⁾	78 95005527 ³⁴⁾	-
50	-	-	-	-	-	66 95005370 ³⁴⁾ 90053914 ³⁵⁾	66 95005520 ³⁴⁾ 90053914 ³⁵⁾	66 95005214 ³⁴⁾ 90053914 ³⁵⁾	66 95005214 ³⁴⁾ 90053914 ³⁵⁾	78 95005527 ³⁴⁾	-
70	-	-	-	-	-	-	-	-	78 95005527 ³⁴⁾	78 95005527 ³⁴⁾	-

Table 49: Cable connector: extension cable for Pt100 sensor, →

Sensor cable for Pt100	Extension cable [mm ²]										
[mm ²]	0,5	-	-	-	-	-	-	-	-	-	-
0,5	18 90062234 ³⁴⁾ 90062233 ³⁵⁾	-	-	-	-	-	-	-	-	-	-

Table 50: Cable connector: extension cable for Pt100 sensor, →

Sensor cable for Pt100 [mm²]	Extension cable [mm²]										
	1,5	2,5	4	6	10	16	25	35	50	70	95
35	-	-	-	-	-	-	-	35 95005172 ³⁴⁾ 90023666 ³⁵⁾	-	-	-
50	-	-	-	-	-	-	-	-	35 95005173 ³⁴⁾ 90023666 ³⁵⁾	-	-
70	-	-	-	-	-	-	-	-	-	35 95005174 ³⁴⁾ 90023666 ³⁵⁾	-
95	-	-	-	-	-	-	-	-	-	-	43 95005188 ³⁴⁾

Cable connector for a shielded electric extension cable
Table 51: Cable connector: shielded electric extension cable, →

Motor lead	Shielded electric extension cable [mm²]									
[mm²]	1,5	2,5	4	6	10	16	25	35	50	70
										
1,5	-	-	-	-	-	-	-	-	-	-
2,5	-	-	-	35 90049387 ³⁶⁾ 90049397 ³⁷⁾	-	43 90049389 ³⁶⁾ 90049399 ³⁷⁾	-	53 90049391 ³⁶⁾ 90049401 ³⁷⁾	-	-
4	-	-	-	35 90049387 ³⁶⁾ 90049397 ³⁷⁾	-	43 90049389 ³⁶⁾ 90049399 ³⁷⁾	-	53 90049391 ³⁶⁾ 90049401 ³⁷⁾	-	-
6	-	-	-	35 90049387 ³⁶⁾ 90049397 ³⁷⁾	-	43 90049389 ³⁶⁾ 90049399 ³⁷⁾	-	53 90049391 ³⁶⁾ 90049401 ³⁷⁾	-	-
10	-	-	-	-	-	43 90049389 ³⁶⁾ 90053910 ³⁷⁾	-	53 90049391 ³⁶⁾ 90049401 ³⁷⁾	66 95005523 ³⁶⁾ 90053914 ³⁷⁾	-
16	-	-	-	-	-	53 95005192 ³⁶⁾ 90053913 ³⁷⁾	-	53 95005192 ³⁶⁾ 90053913 ³⁷⁾	66 95005523 ³⁶⁾ 90053914 ³⁷⁾	66 95005523 ³⁶⁾ 90053914 ³⁷⁾
25	-	-	-	-	-	-	-	66 95005214 ³⁶⁾ 90053914 ³⁷⁾	66 95005523 ³⁶⁾ 90053914 ³⁷⁾	66 95005523 ³⁶⁾ 90053914 ³⁷⁾

36 MVB - material No. for cable connector, extended and sealed during manufacture

37 OVB - material No. for cable connector, not connected and supplied but not fitted

Table 52: Cable connector: shielded electric extension cable, + or + →

Motor lead [mm²]	Shielded electric extension cable [mm²]									
	1,5	2,5	4	6	10	16	25	35	50	70
+ or +										
1,5	-	-	-	-	-	-	-	-	-	-
2,5	-	-	-	35 95005165 ³⁶⁾ 90039349 ³⁷⁾	-	-	-	-	-	-
4	-	-	-	-	-	43 90049390 ³⁶⁾ 90049400 ³⁷⁾	-	-	-	-
6	-	-	-	-	-	43 90049390 ³⁶⁾ 90049400 ³⁷⁾	-	53 90049392 ³⁶⁾ 90049402 ³⁷⁾	66 90033248 ³⁶⁾ 90055737 ³⁷⁾	66 90033248 ³⁶⁾ 90055737 ³⁷⁾
10	-	-	-	-	-	-	-	66 90039412 ³⁶⁾ 90053905 ³⁷⁾	66 90039412 ³⁶⁾ 90053905 ³⁷⁾	66 90033248 ³⁶⁾ 90055737 ³⁷⁾
16	-	-	-	-	-	-	-	66 90039361 ³⁶⁾ 90053906 ³⁷⁾	66 90036538 ³⁶⁾ 90055738 ³⁷⁾	66 90036538 ³⁶⁾ 90055738 ³⁷⁾
25	-	-	-	-	-	-	-	-	78 90039701 ³⁶⁾ 90053908 ³⁷⁾	78 90039701 ³⁶⁾ 90053908 ³⁷⁾
35	-	-	-	-	-	-	-	-	-	-
50	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-

Table 53: Cable connector: shielded electric extension cable, →

Motor lead [mm²]	Shielded electric extension cable [mm²]									
	1,5	2,5	4	6	10	16	25	35	50	70
1,5	-	-	-	-	-	-	-	-	-	-
2,5	-	-	-	28 90049385 ³⁶⁾ 95005106 ³⁷⁾	-	43 90049389 ³⁶⁾ 90049399 ³⁷⁾	-	-	-	-
4	-	-	-	28 90049385 ³⁶⁾ 95005106 ³⁷⁾	-	43 90049389 ³⁶⁾ 90049399 ³⁷⁾	-	-	-	-
6	-	-	-	28 90049385 ³⁶⁾ 95005106 ³⁷⁾	-	43 90049389 ³⁶⁾ 90049399 ³⁷⁾	-	53 90049391 ³⁶⁾ 90049401 ³⁷⁾	-	-
10	-	-	-	-	-	43 90049389 ³⁶⁾ 90049399 ³⁷⁾	-	53 90049391 ³⁶⁾ 90049401 ³⁷⁾	66 95005523 ³⁶⁾ 90053914 ³⁷⁾	-
16	-	-	-	-	-	43 90049389 ³⁶⁾ 90049399 ³⁷⁾	-	53 90049391 ³⁶⁾ 90049401 ³⁷⁾	66 95005370 ³⁶⁾ 90053914 ³⁷⁾	66 95005214 ³⁶⁾ 90053914 ³⁷⁾
25	-	-	-	-	-	-	-	53 90049391 ³⁶⁾ 90049401 ³⁷⁾	66 95005520 ³⁶⁾ 90053914 ³⁷⁾	78 95005527 ³⁶⁾ 05244394 ³⁷⁾

Motor lead	Shielded electric extension cable [mm ²]									
[mm ²]	1,5	2,5	4	6	10	16	25	35	50	70
										
35	-	-	-	-	-	-	-	53 90049391 ⁽³⁶⁾ 90049401 ⁽³⁷⁾	66 95005214 ⁽³⁶⁾ 90053914 ⁽³⁷⁾	78 95005527 ⁽³⁶⁾ 05244395 ⁽³⁷⁾
50	-	-	-	-	-	-	-	-	66 95005214 ⁽³⁶⁾ 90053914 ⁽³⁷⁾	78 95005527 ⁽³⁶⁾ 05244396 ⁽³⁷⁾
70	-	-	-	-	-	-	-	-	78 95005527 ⁽³⁶⁾ 05244396 ⁽³⁷⁾	78 95005527 ⁽³⁶⁾ 05244397 ⁽³⁷⁾

Cable clips

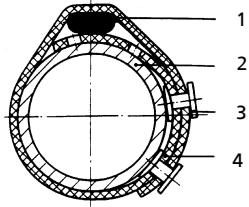
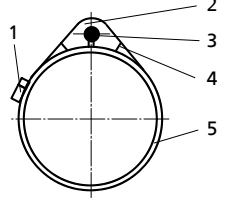
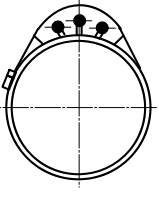
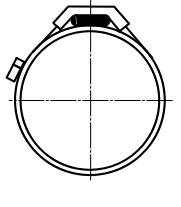
Application

- Securing electric cables at the riser / discharge pipe
- Suitable for use in normal or slightly aggressive water
- Versions for use in more aggressive water (e.g. seawater) on request

Table 54: Selection table

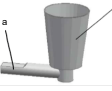
Electric cable			Cable clip size	Cable guard	Mat. No.	Weight	Required strap length L [mm] per cable clip for the following column pipes													
Type	Number x Ø of cores						[kg]	G 1 1/4 / DN 32	G 1 1/2 / DN 40	G 2 / DN 50	DN 65	G 3 / DN 80	G 4 / DN 100	G 5 / DN 125	G 6 / DN 150	DN 175	G 8 / DN 200	G 10 / DN 250	G 12 / DN 300	G 14 / DN 350
	[mm]																			
Flat	3 x 1.5 to 3 x 6.0	4 x 1.5 to 4 x 6.0	1	-	01088095	0.04	270	280	320	360	400	450	500	600	-	-	-	-	-	-
	3 x 10 to 3 x 70	4 x 10 to 4 x 70	11	✓	On request	0,01	-	-	350	400	450	550	650	800	950	1100	1300	1500	1700	1900
Round	3 x 1.5 to 3 x 6.0	4 x 1.5 to 4 x 6.0	1	-	01088095	0.04	270	280	320	360	400	450	500	600	-	-	-	-	-	-
	3 x 10	4 x 10	2	✓	On request	0,08	-	-	350	400	450	550	650	800	950	1100	1300	1500	1700	1900
	3 x 16 3 x 25	4 x 16	3a	-	On request	0,19	-	-	350	400	450	550	650	800	950	1100	1300	1500	1700	1900
	3 x 35 3 x 50	4 x 25 4 x 35	3	-	On request	0,19	-	-	-	450	500	600	700	850	1000	1150	1350	1550	1750	1950
	3 x 70	4 x 50	3b	-	On request	0,19	-	-	-	450	500	600	700	850	1000	1150	1350	1550	1750	1950
	3 x 95	4 x 70 4 x 95	4	-	On request	0,55	-	-	-	-	650	750	900	1050	1200	1400	1600	1800	2000	
	3 cables 1 x 50		7	-	On request	0,6	-	-	-	-	-	700	800	950	1100	1250	1450	1650	1850	2050
	3 cables 1 x 70		8	-	On request	0,6	-	-	-	-	-	700	800	950	1100	1250	1450	1650	1850	2050
	3 cables 1 x 95		6	-	On request	0,6	-	-	-	-	-	700	800	950	1100	1250	1450	1650	1850	2050
	3 cables 1 x 120		9	✓	On request	0,6	-	-	-	-	-	700	800	950	1100	1250	1450	1650	1850	2050
	3 cables 1 x 150		9	✓	On request	0,6	-	-	-	-	-	700	800	950	1100	1250	1450	1650	1850	2050

Table 55: Cable clip sizes

Size 1	Sizes 2, 3, 3a, 3b, 4	Sizes 6 to 9	Size 11
			
1 = electric cable 2 = riser 3 = plastic stud 4 = rubber strap	1 = screw band with tension screw (re-usable) 2 = cable guard 3 = electric cable 4 = metal strap 5 = riser		

Motor refill kit

Table 56: Overview of motor refill kit

	Description	Mat. No.	Weight
			[kg]
	Fill check kit for checking and topping up the motor fill after prolonged storage for UMA 150, UMA-S 150, UMA 200, UMA-S 200, UMA 250, UMA-S 250 motors	90066762	0.25
	Refill kit for checking and topping up the motor fill after prolonged storage for UMA 300 motor	Pipe (a)	0.01
		Filling funnel (b)	0.05

Monitoring and automation

Table 57: Selection table

Description		Mat. No.	Weight [kg/m]
Dry running protection device (supply voltage 220 V / 240 V) for			
	Semi-automatic operation (1 relay, 1 pushbutton + 2 electrodes)	90009553	2
	Fully automatic operation (1 relay + 3 electrodes)	90009554	2
	Plus 1 control cable, blue, 1 x 1.5 mm ² for each electrode; values indicated per metre	01096713	0.1
Immersion electrodes			
	Set of electrodes: 3 electrodes made of stainless steel	40980055	0.3
	Set of electrodes: 1 electrode made of stainless steel	40980056	0.1
Pressure controller , insulating enclosure (membrane type), 1 to 8 bar, with pressure connection G 3/8		01151586	1
Pt100 temperature monitoring equipment for use in submersible motors to protect the windings from overheating. Comprising:			
Pt100 temperature sensor with 10 m cable 4 x 0.5 mm ² for motor:	UMA 150 material variants E, C	01532593	1.1
	UMA 150 material variant D	01532594	1.1
	UMA 200, UMA 250, material variants G, C	90063006	1.1
	UMA 200, UMA 250, material variants D	On request	-
Display / switchgear (supply voltage V = 110/220 V)		90064446	0.45
Control cable, blue, round 4 x 0.5 mm ²		Per metre	01049403
			0.28

Frequency inverter and voltage filter

Table 58: Selection table

Motor UMA-S								Frequency inverter PumpDrive R				dv/dt filter				Sine filter			
Material variant				Size	P _N		I _N	Enclosure	Mat. No.	Designation	I _A	Enclosure	Mat. No.	Designation	I _{NFilter}	Enclosure	Mat. No.	Designation	I _{NFilter}
G	E	C	D		[kW]	[Hp]													
Mat. No.	Mat. No.	Mat. No.	Mat. No.		[kW]	[Hp]	[A]		[A]	[A]									
-	01639317	01732881	01732882	UMA-S 150 7/42	4,5	6	8	IP20	48229690	PumpDrive R KSB202 004K00C	10	IP00	01664325	130B2835	33	IP00	01664341	130B2411	13
-	01639317	01732881	01732882	UMA-S 150 7/42	5,5	7	10	IP20	48229692	PumpDrive R KSB202 005K50C	13	IP00	01664325	130B2835	33	IP00	01664341	130B2411	13
-	01639317	01732881	01732882	UMA-S 150 7/42	7,5	10	13	IP20	48229694	PumpDrive R KSB202 007K50C	16	IP00	01664325	130B2835	33	IP00	01664342	130B2412	18
-	01584165	01732855	01732856	UMA-S 150 18/42	9,3	12	18	IP20	48229696	PumpDrive R KSB202 011K00C	24	IP00	01664325	130B2835	33	IP00	01664343	130B2413	28,5
-	01584165	01732855	01732856	UMA-S 150 18/42	11	15	20	IP20	48229696	PumpDrive R KSB202 011K00C	24	IP00	01664325	130B2835	33	IP00	01664343	130B2413	28,5
-	01584165	01732855	01732856	UMA-S 150 18/42	13	17	23	IP20	48229698	PumpDrive R KSB202 015K00C	32	IP00	01664325	130B2835	33	IP00	01664344	130B2281	36
-	01584165	01732855	01732856	UMA-S 150 18/42	15	20	26	IP20	48229698	PumpDrive R KSB202 015K00C	32	IP00	01664325	130B2835	33	IP00	01664344	130B2281	36
-	01584165	01732855	01732856	UMA-S 150 18/42	18,5	25	32	IP20	48229700	PumpDrive R KSB202 018K50C	37,5	IP00	01664326	130B2838	68	IP00	01664345	130B2282	46,5
-	01584166	01732877	01732878	UMA-S 150 37/42	22	29	39	IP20	48229702	PumpDrive R KSB202 022K00C	44	IP00	01664326	130B2838	68	IP00	01664345	130B2282	46,5
-	01584166	01732877	01732878	UMA-S 150 37/42	26	35	46	IP20	48229704	PumpDrive R KSB202 030K00C	61	IP00	01664326	130B2838	68	IP00	01664346	130B3179	86
-	01584166	01732877	01732878	UMA-S 150 37/42	30	40	54	IP20	48229704	PumpDrive R KSB202 030K00C	61	IP00	01664326	130B2838	68	IP00	01664346	130B3179	86
-	01584197	01732879	01732880	UMA-S 150 37/42	37	49	72	IP20	48229708	PumpDrive R KSB202 045K00C	90	IP00	01674691	130B2844	133	IP00	01664346	130B3179	86
-	01639317	01732881	01732882	UMA-S 150 7/42	4,5	6	8	IP55	48229691	PumpDrive R KSB202 004K00W	10	IP54	01664338	130B2837	34	IP54	01664347	192H3025	17
-	01639317	01732881	01732882	UMA-S 150 7/42	5,5	7	10	IP55	48229693	PumpDrive R KSB202 005K50W	13	IP54	01664338	130B2837	34	IP54	01664347	192H3025	17
-	01639317	01732881	01732882	UMA-S 150 7/42	7,5	10	13	IP55	48229695	PumpDrive R KSB202 007K50W	16	IP54	01664338	130B2837	34	IP54	01664347	192H3025	17
-	01584165	01732855	01732856	UMA-S 150 18/42	9,3	12	18	IP55	48229697	PumpDrive R KSB202 011K00W	24	IP54	01664338	130B2837	34	IP54	01664348	192H3027	24,2
-	01584165	01732855	01732856	UMA-S 150 18/42	11	15	20	IP55	48229697	PumpDrive R KSB202 011K00W	24	IP54	01664338	130B2837	34	IP54	01664348	192H3027	24,2
-	01584165	01732855	01732856	UMA-S 150 18/42	13	17	23	IP55	48229699	PumpDrive R KSB202 015K00W	32	IP54	01664338	130B2837	34	IP54	01664349	192H3028	32
-	01584165	01732855	01732856	UMA-S 150 18/42	15	20	26	IP55	48229699	PumpDrive R KSB202 015K00W	32	IP54	01664338	130B2837	34	IP54	01664349	192H3028	32
-	01584165	01732855	01732856	UMA-S 150 18/42	18,5	25	32	IP55	48229701	PumpDrive R KSB202 018K50W	37,5	IP54	01664339	130B2840	56	IP54	01664350	192H3010	37,5

Motor UMA-S								Frequency inverter PumpDrive R				dv/dt filter					Sine filter			
Material variant				Size	P _N		I _N	Enclosure	Mat. No.	Designation	I _A	Enclosure	Mat. No.	Designation	I _{NFilter}	Enclosure	Mat. No.	Designation	I _{NFilter}	
G	E	C	D		[kW]	[Hp]	[A]				[A]				[A]				[A]	
Mat. No.	Mat. No.	Mat. No.	Mat. No.																	[A]
-	01584166	01732877	01732878	UMA-S 150 37/42	22	29	39	IP55	48229703	PumpDrive R KSB202 022K00W	44	IP54	01664339	130B2840	56	IP54	01664351	192H3011	46,2	
-	01584166	01732877	01732878	UMA-S 150 37/42	26	35	46	IP55	48229705	PumpDrive R KSB202 030K00W	61	IP54	01664340	130B2843	83	IP54	01664352	192H3035	61	
-	01584166	01732877	01732878	UMA-S 150 37/42	30	40	54	IP55	48229705	PumpDrive R KSB202 030K00W	61	IP54	01664340	130B2843	83	IP54	01664352	192H3035	61	
-	01584197	01732879	01732880	UMA-S 150 37/42	37	49	72	IP55	48229709	PumpDrive R KSB202 045K00W	90	IP54	01664340	130B2843	83	IP54	01674692	192H3037	115	
01738152	-	01738156	01738170	UMA-S 200 75/42	45	60	74	IP20	48229708	PumpDrive R KSB202 045K00C	90	IP00	01674691	130B2844	133	IP00	01664346	130B3179	86	
01738152	-	01738156	01738170	UMA-S 200 75/42	55	74	91	IP20	48229710	PumpDrive R KSB202 055K00C	106	IP00	01674691	130B2844	133	IP00	01733111	130B3182	135	
01738152	-	01738156	01738170	UMA-S 200 75/42	67	90	112	IP20	48229712	PumpDrive R KSB202 075K00C	147	IP00	01674691	130B2844	133	IP00	01733111	130B3182	135	
01738152	-	01738156	01738170	UMA-S 200 75/42	75	100	128	IP20	48229714	PumpDrive R KSB202 090K00C	177	IP00	01733056	130B2847	236	IP00	01733112	130B3184	195	
01738153	-	01738167	01738171	UMA-S 200 100/42	75	100	129	IP20	48229714	PumpDrive R KSB202 090K00C	177	IP00	01733056	130B2847	236	IP00	01733112	130B3184	195	
01738153	-	01738167	01738171	UMA-S 200 100/42	83	110	143	IP20	48229714	PumpDrive R KSB202 090K00C	177	IP00	01733056	130B2847	236	IP00	01733112	130B3184	195	
01738153	-	01738167	01738171	UMA-S 200 100/42	93	125	163	IP20	48229716	PumpDrive R KSB202 110K00C	212	IP00	01733056	130B2847	236	IP00	01733112	130B3184	195	
01738153	-	01738167	01738171	UMA-S 200 100/42	100	135	178	IP20	48229716	PumpDrive R KSB202 110K00C	212	IP00	01733056	130B2847	236	IP00	01733112	130B3184	195	
05148497	-	05148498	05148551	UMA-S 200 130/42	110	145	186	IP20	01733781	PumpDrive R KSB202 132K00C	260	IP00	01733056	130B2847	236	IP00	01733113	130B3186	308	
05148497	-	05148498	05148551	UMA-S 200 130/42	130	175	224	IP20	01733783	PumpDrive R KSB202 160K00C	315	IP00	01733056	130B2847	236	IP00	01733113	130B3186	308	
01738152	-	01738156	01738170	UMA-S 200 75/42	45	60	74	IP55	48229709	PumpDrive R KSB202 045K00W	90	IP54	01733108	130B2846	133	IP54	01674692	192H3037	115	
01738152	-	01738156	01738170	UMA-S 200 75/42	55	74	91	IP55	48229711	PumpDrive R KSB202 055K00W	106	IP54	01733108	130B2846	133	IP54	01674692	192H3037	115	
01738152	-	01738156	01738170	UMA-S 200 75/42	67	90	112	IP55	48229713	PumpDrive R KSB202 075K00W	147	IP54	01733108	130B2846	133	IP54	01674692	192H3037	115	
01738152	-	01738156	01738170	UMA-S 200 75/42	75	100	128	IP55	48229715	PumpDrive R KSB202 090K00W	177	IP23	01733109	130B2848	236	IP23	01733115	130B9027	195	
01738153	-	01738167	01738171	UMA-S 200 100/42	75	100	129	IP55	48229715	PumpDrive R KSB202 090K00W	177	IP23	01733109	130B2848	236	IP23	01733115	130B9027	195	
01738153	-	01738167	01738171	UMA-S 200 100/42	83	110	143	IP55	48229715	PumpDrive R KSB202 090K00W	177	IP23	01733109	130B2848	236	IP23	01733115	130B9027	195	
01738153	-	01738167	01738171	UMA-S 200 100/42	93	125	163	IP55	48229717	PumpDrive R KSB202 110K00W	212	IP23	01733109	130B2848	236	IP23	01733115	130B9027	195	

Motor UMA-S								Frequency inverter PumpDrive R				dv/dt filter				Sine filter			
Material variant				Size	P _N		I _N	Enclosure	Mat. No.	Designation	I _A	Enclosure	Mat. No.	Designation	I _{NFilter}	Enclosure	Mat. No.	Designation	I _{NFilter}
G	E	C	D		[kW]	[Hp]													
Mat. No.	Mat. No.	Mat. No.	Mat. No.																
01738153	-	01738167	01738171	UMA-S 200 100/42	100	135	178	IP55	48229717	PumpDrive R KSB202 110K00W	212	IP23	01733109	130B2848	236	IP23	01733115	130B9027	195
05148497	-	05148498	05148551	UMA-S 200 130/42	110	145	186	IP54	01733782	PumpDrive R KSB202 132K00W	260	IP23	01733109	130B2848	236	IP23	01733116	130B3187	308
05148497	-	05148498	05148551	UMA-S 200 130/42	130	175	224	IP54	01733784	PumpDrive R KSB202 160K00W	315	IP23	01733109	130B2848	236	IP23	01733116	130B3187	308
05147847	-	05147848	05147849	UMA-S 250 200/42	185	246	358	IP20	01839750	PumpDrive R KSB202 200K00C	395	IP00	01733107	130B2849	360	IP00	01845293	130B3188	382
05147847	-	05147848	05147849	UMA-S 250 200/42	200	266	389	IP20	01839789	PumpDrive R KSB202 250K00C	480	IP00	05146486	130B2851	494	IP00	05146604	130B3191	496
05147882	-	05147883	05147901	UMA-S 250 250/42	230	306	447	IP20	01839789	PumpDrive R KSB202 250K00C	480	IP00	05146486	130B2851	494	IP00	05146604	130B3191	496
05147882	-	05147883	05147901	UMA-S 250 250/42	250	333	493	IP20	05117684	PumpDrive R KSB202 315K00C	600	IP00	05146488	130B2853	660	IP00	05146622	130B3193	601
05147847	-	05147848	05147849	UMA-S 250 200/42	185	246	358	IP54	01839790	PumpDrive R KSB202 200K00W	395	IP23	01733110	130B2850	360	IP23	01845295	130B3189	382
05147847	-	05147848	05147849	UMA-S 250 200/42	200	266	389	IP54	01839792	PumpDrive R KSB202 250K00C	480	IP23	05146487	130B2852	494	IP23	05146605	130B3192	496
05147882	-	05147883	05147901	UMA-S 250 250/42	230	306	447	IP54	01839792	PumpDrive R KSB202 250K00C	480	IP23	05146487	130B2852	494	IP23	05146605	130B3192	496
05147882	-	05147883	05147901	UMA-S 250 250/42	250	333	493	IP54	05117687	PumpDrive R KSB202 315K00W	600	IP23	05146597	130B2854	660	IP23	05146623	130B3194	601



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